完整逐轮曲线 · 方法代码说明

聚合与基础 LoRA 诊断 = first experiment batch · 组会后续实验 = second experiment batch

Each color is one seed (7, 13, 21, 42, 87). Solid = trained model; dashed = EMA readout. All rounds are shown without smoothing.

聚合与基础 LoRA 诊断

central_iid_both
Central IID batches; train both LoRA factors

central_pooled_both
Central pooled cohort; train both factors

central_serial_alt
Central serial client blocks; alternate A/B

central_serial_both
Central serial client blocks; train A/B

fed_both_exact
Federated A/B with exact residual correction

fed_both_mean
Federated A/B with separate factor means

fed_ffa_b
Federated frozen A; train B only

online_pcla_contributors
Online PCLA; contributor denominator C

online_replay_contributors
Online random-rank replay; denominator C

trace_pcla_contributors
Saved PCLA support trace; denominator C

trace_pcla_function_norm_matched
Saved PCLA trace; norm-matched scale G

trace_random_contributors
Saved random support trace; denominator C

trace_random_function_norm_matched
Saved random trace; norm-matched scale G

trace_replay_random_rank_contributors
Saved replay support trace; denominator C

trace_replay_random_rank_function_norm_matched
Saved replay trace; norm-matched scale G

组会后续实验

D_agnews_central_iid_both
AG_NEWS central IID; train both factors

D_agnews_fed_both_exact
AG_NEWS federated A/B; exact residual

D_agnews_fed_both_mean
AG_NEWS federated A/B; factor means

D_agnews_full_alt
AG_NEWS all ranks; alternate A/B

D_agnews_online_C
AG_NEWS online PCLA; denominator C

D_agnews_pcla
AG_NEWS PCLA

D_agnews_replay
AG_NEWS same layers/counts, random rank IDs

C_central_full_alt_r30
Central, all ranks; alternate A/B; 30 rounds

C_central_pcla_r30
Central PCLA; 30 rounds

C_central_random_rank_r30
Central random layers and ranks; 30 rounds

C_central_replay_random_rank_r30
Central same layers/counts, random rank IDs

L1_fed_alt_r100
All ranks; alternate A/B; 100 rounds

L2_pcla_r100
PCLA; 100 rounds; EMA from read-only replay

L3_fed_both_mean_r100
Federated A/B factor means; 100 rounds

P3_fed_sa_r30
FedSA-style shared A and client-specific B

组会后续实验

R_iid_ffa
Upstream FFA-LoRA; IID split

R_iid_normal
Upstream FedIT; IID split

R_iid_ours
Upstream FedEx with residual transpose; IID

R_iid_ours_fixed
Upstream FedEx without transpose; IID

R_noniid_ffa
Upstream FFA-LoRA; non-IID split

R_noniid_normal
Upstream FedIT; non-IID split

R_noniid_ours
Upstream FedEx with transpose; non-IID

R_noniid_ours_fixed
Upstream FedEx without transpose; non-IID

T1_fed_alt_r100_slr05
All ranks; server step size 0.5; 100 rounds

T2_fed_alt_r100_strat
All ranks; stratified clients; 100 rounds

V1_fed_svd_refactor_r30
Exact A/B aggregation, then SVD refactor

V2_alt_svd_refactor_r30
Alternate A/B, then SVD refactor

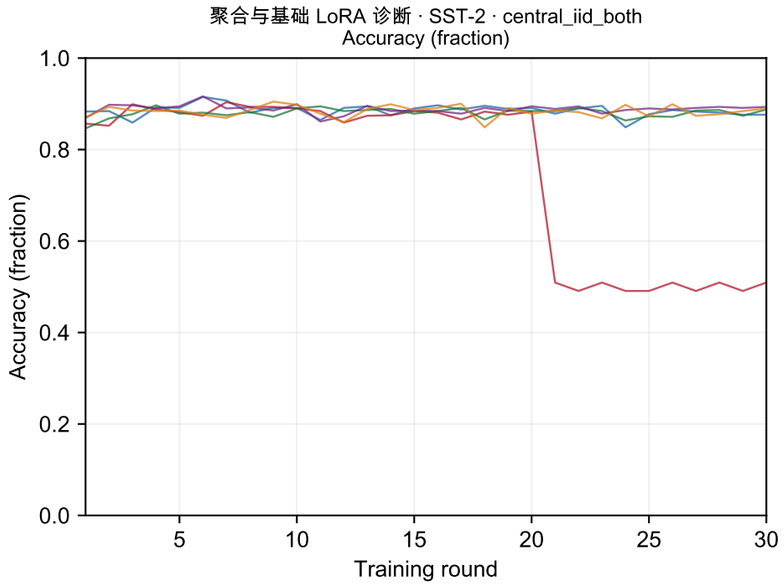
V3_ffa_svd_basis_r30
Frozen pretrained singular-vector basis

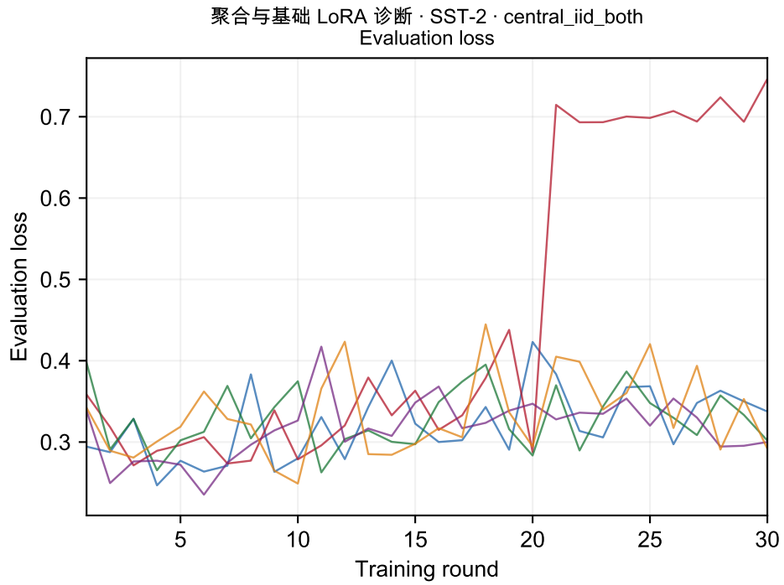
V4_pcla_svd
PCLA with SVD refactor each round

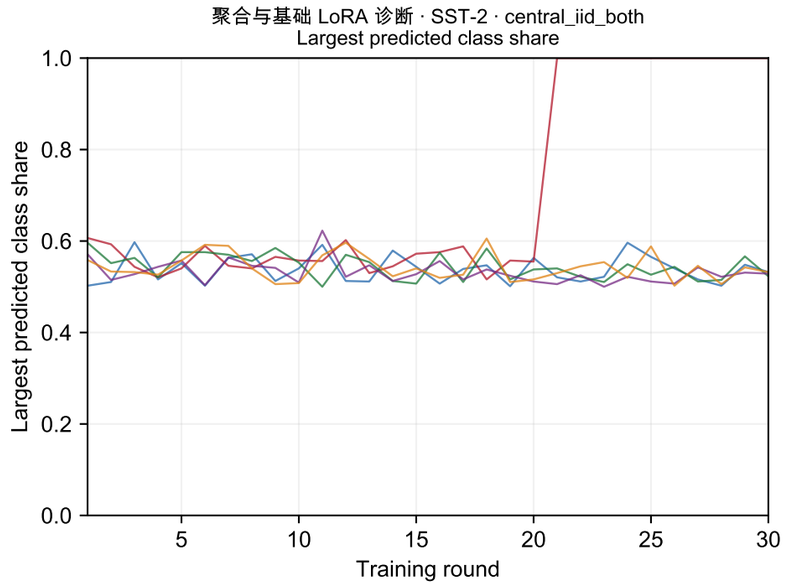
V5_replay_svd
Same layers/counts, random ranks, with SVD

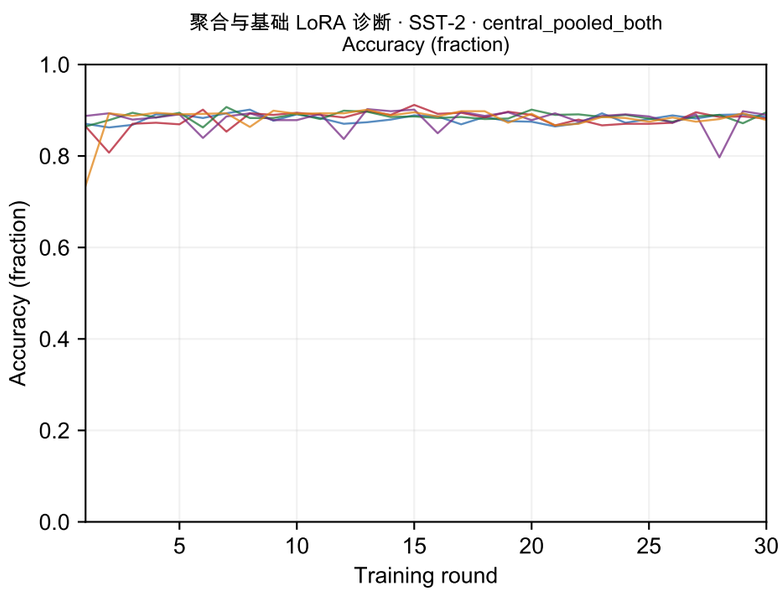
C = averaging by contributing client count; G = function-norm-matched scaling. Counts for every class are in round_curves_public.csv.

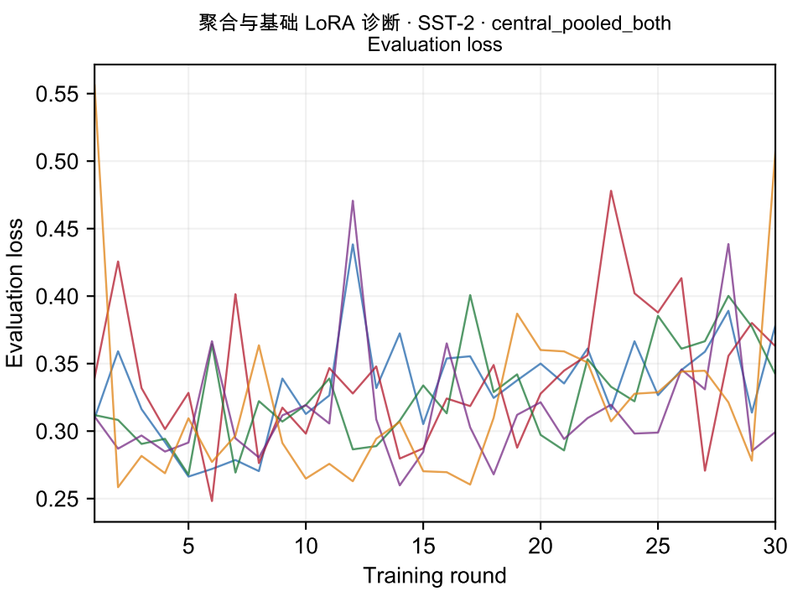
Full round trajectories · methods 1–2 of 45

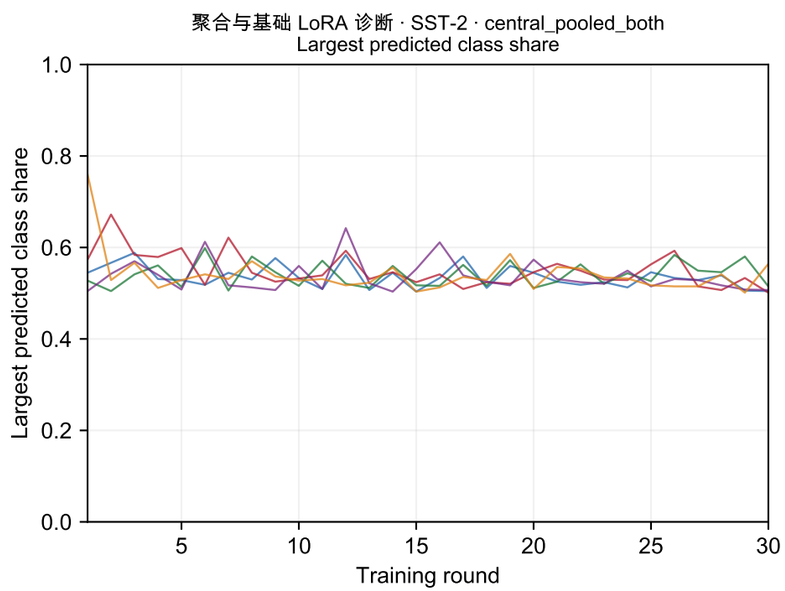


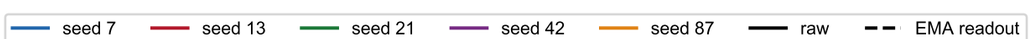




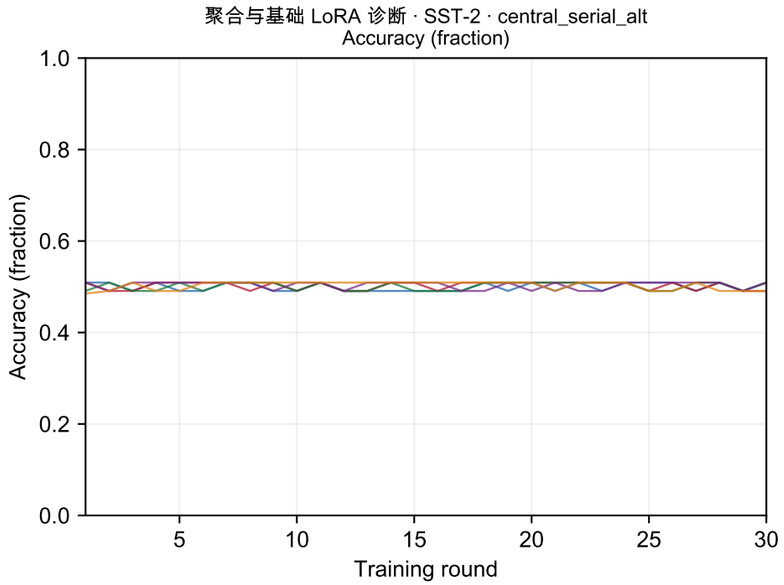


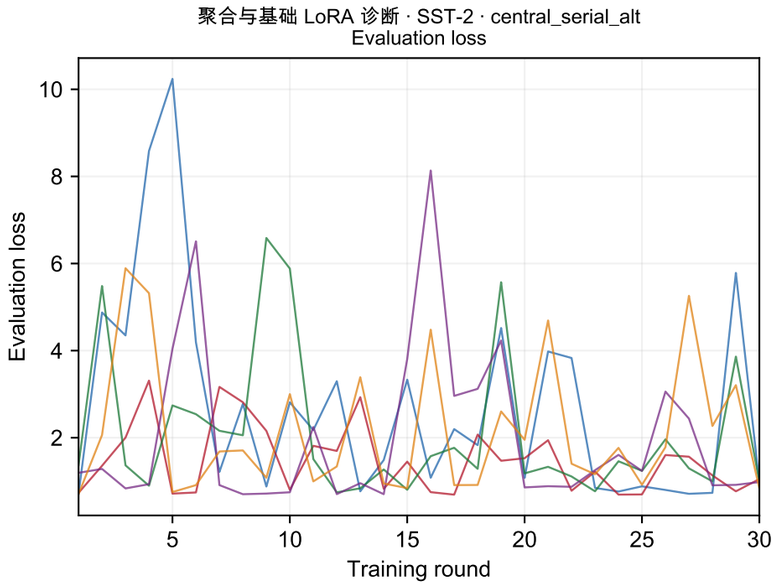


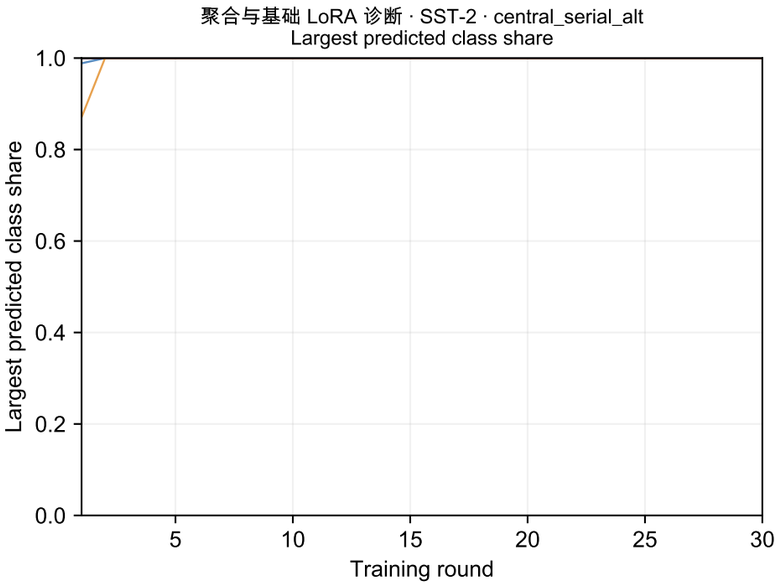


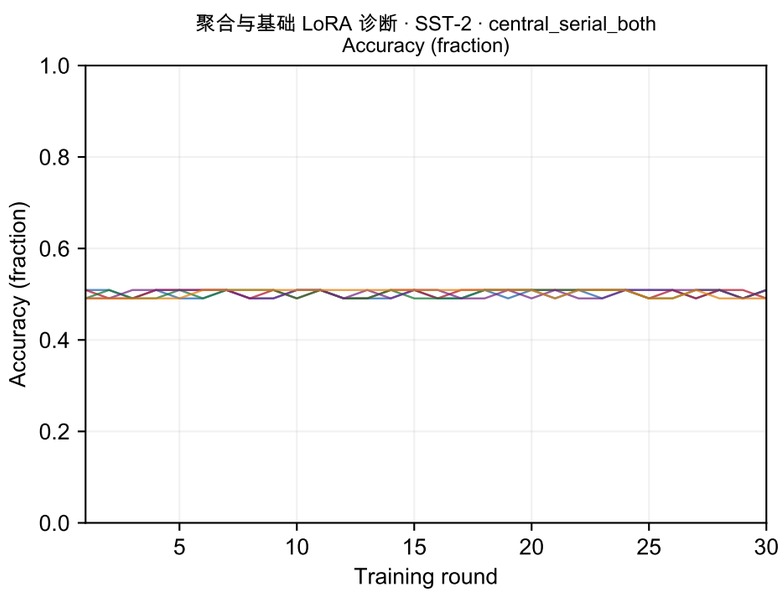


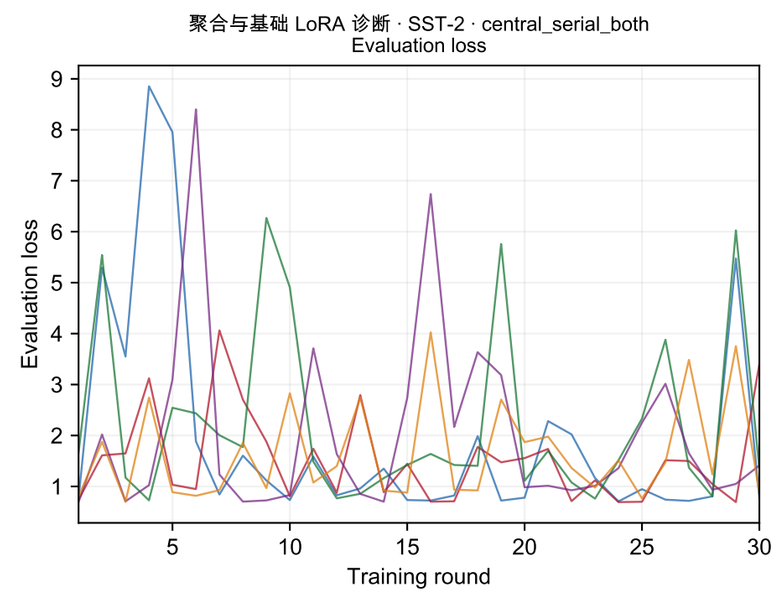
Full round trajectories · methods 3–4 of 45

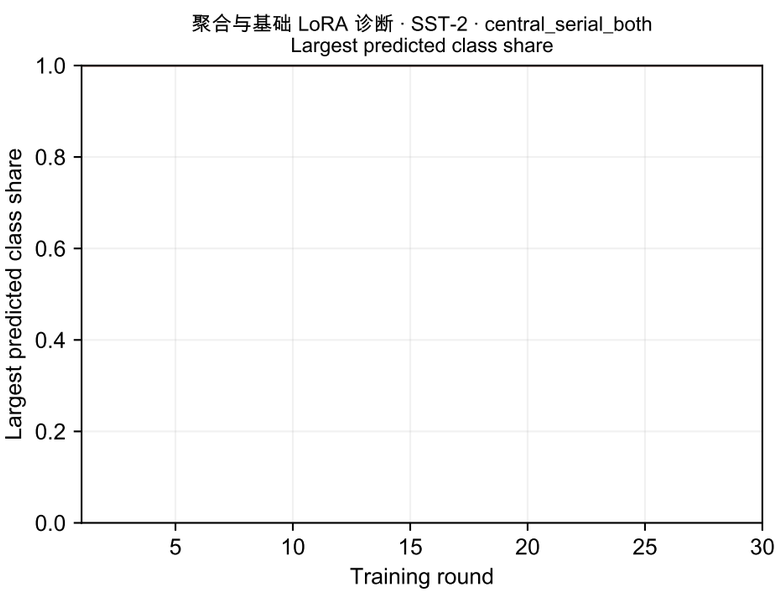


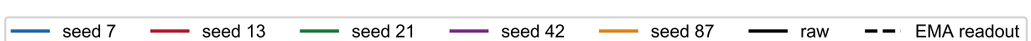




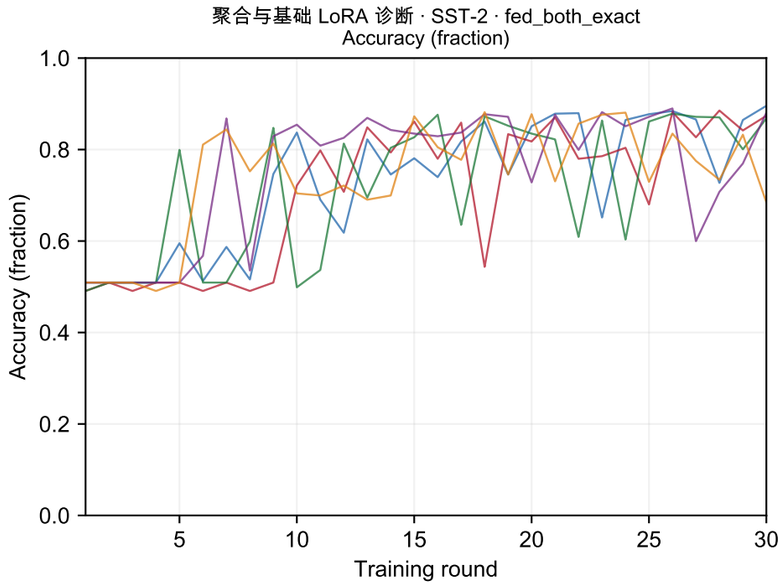


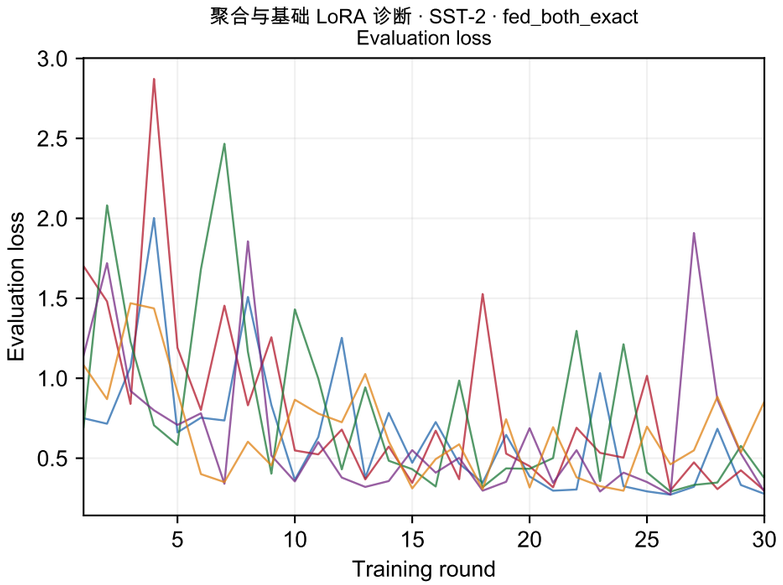


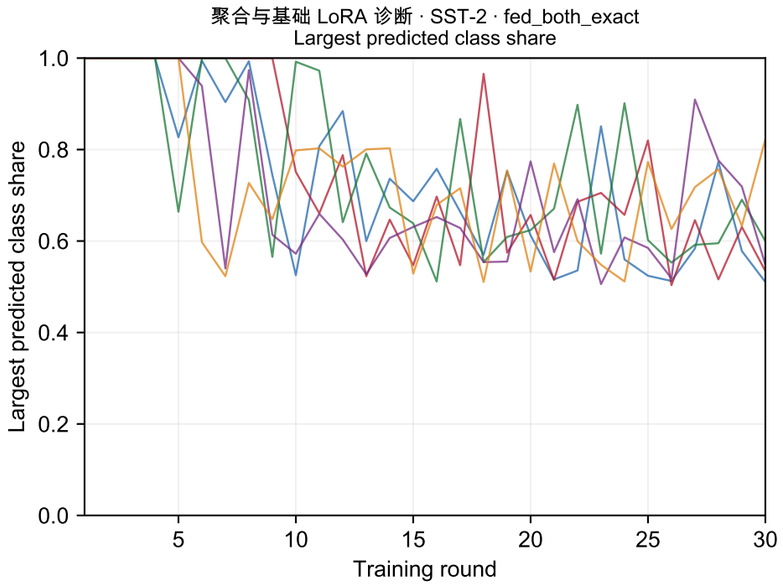


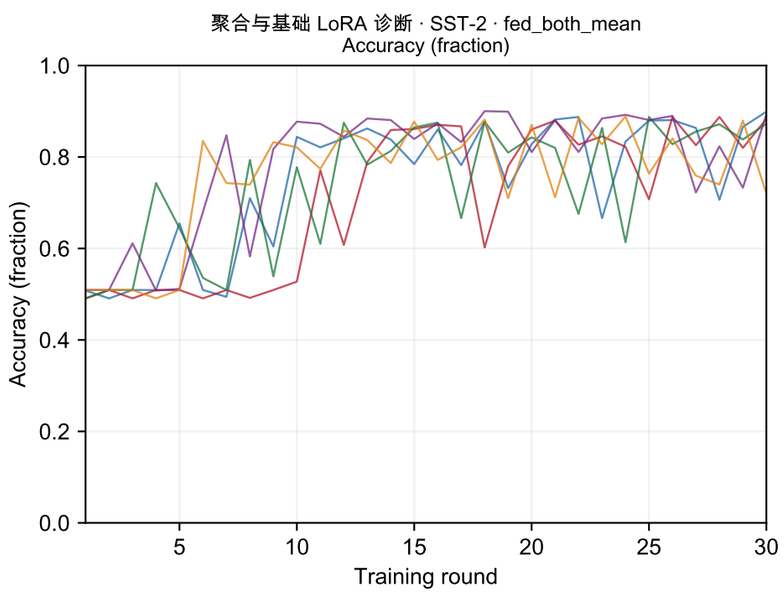


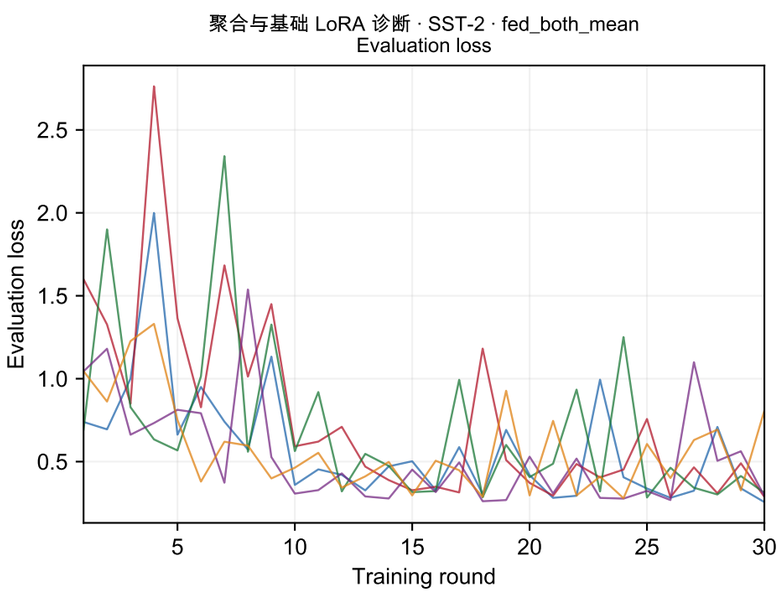
Full round trajectories · methods 5–6 of 45

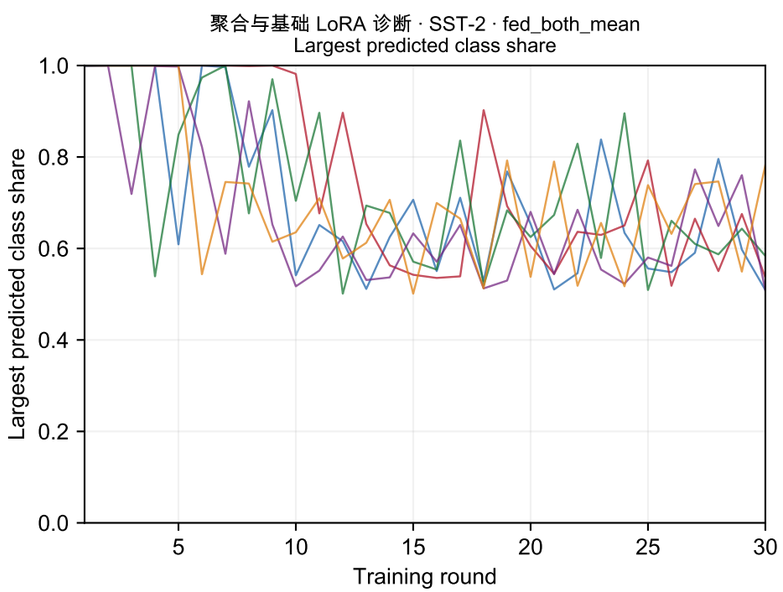


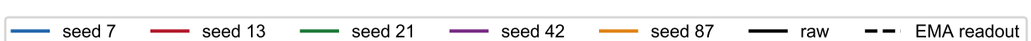




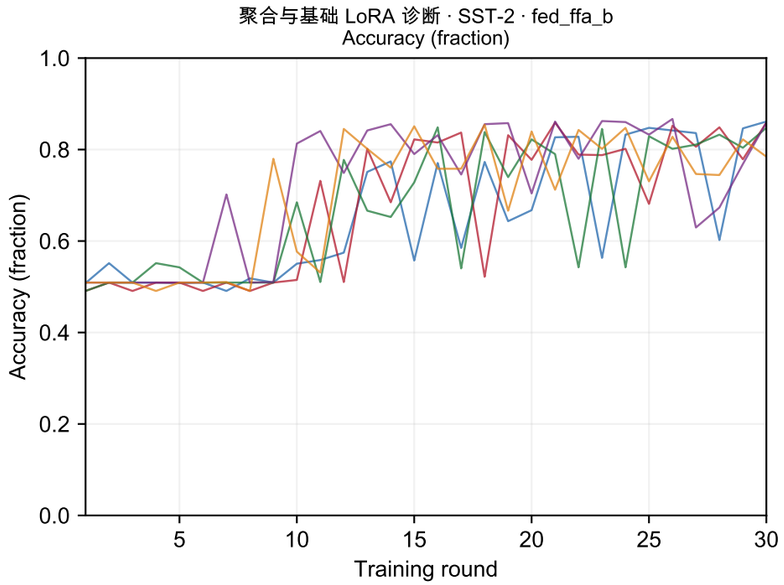


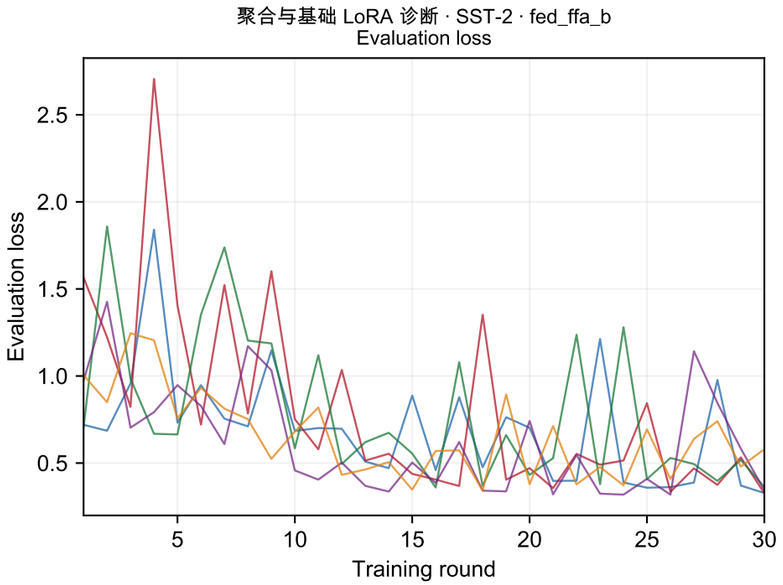


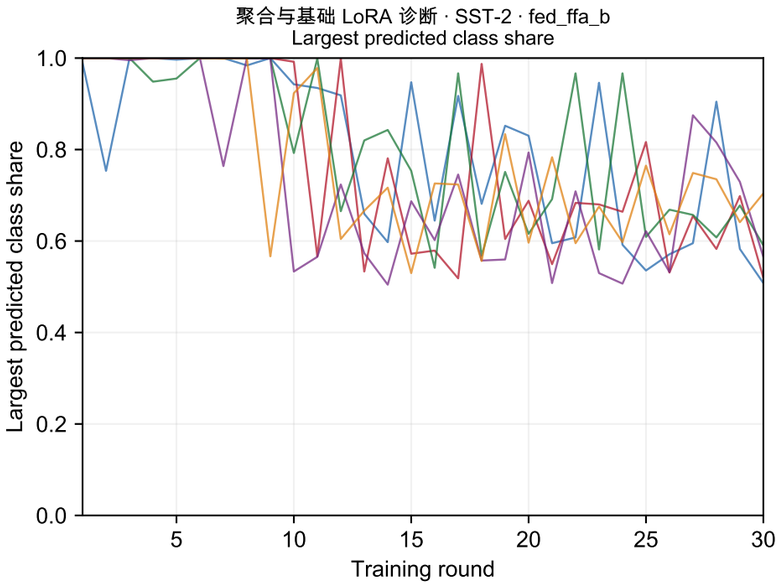


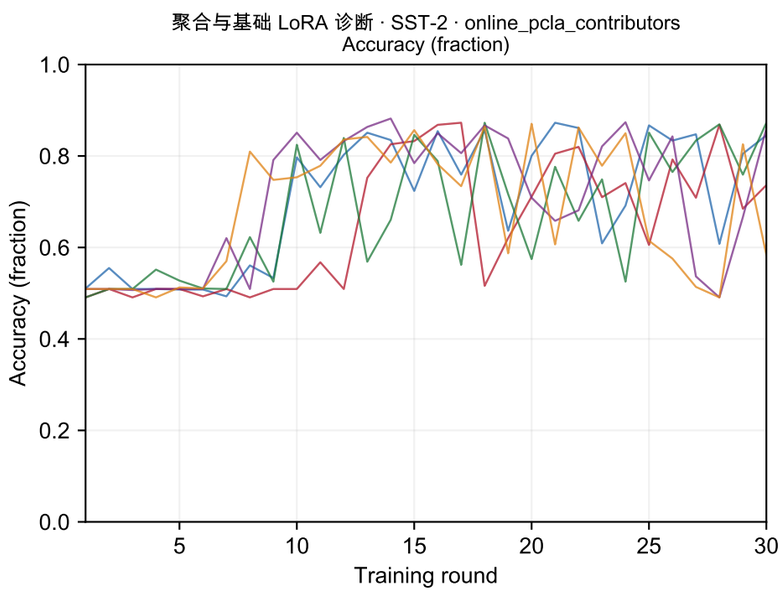


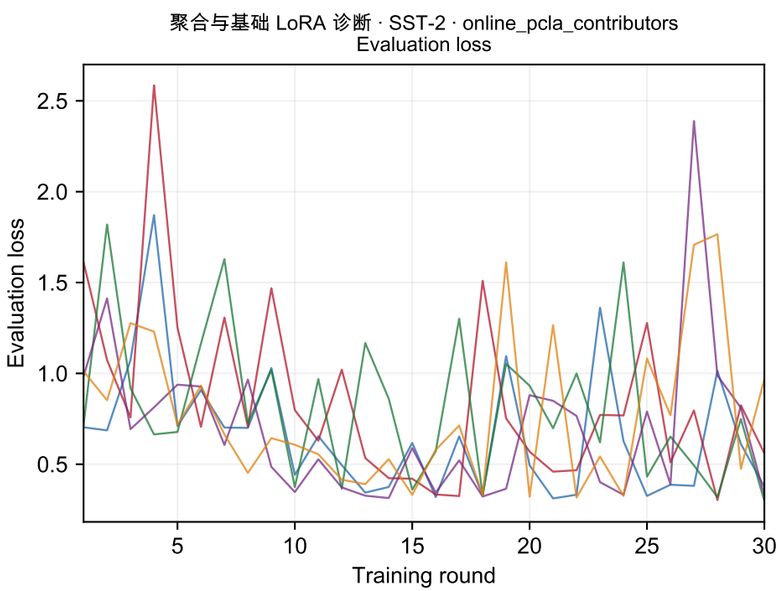
Full round trajectories · methods 7–8 of 45

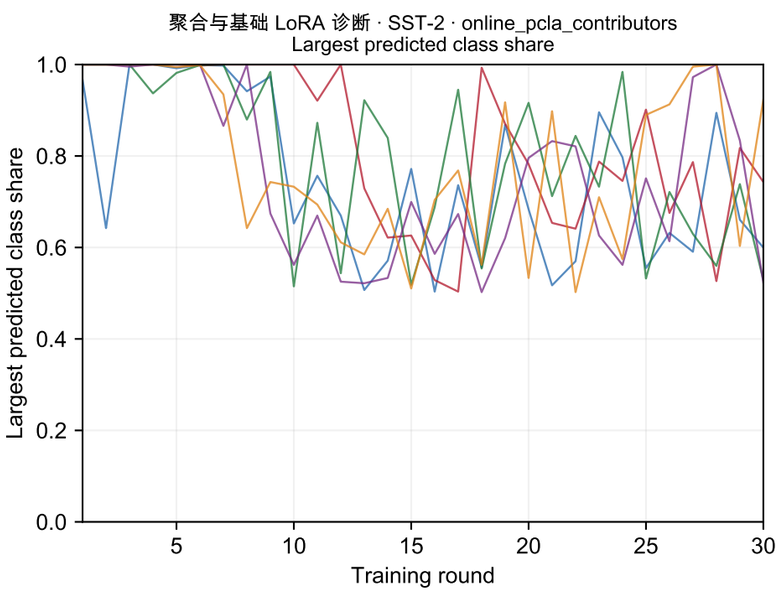


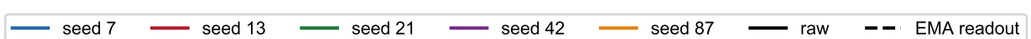




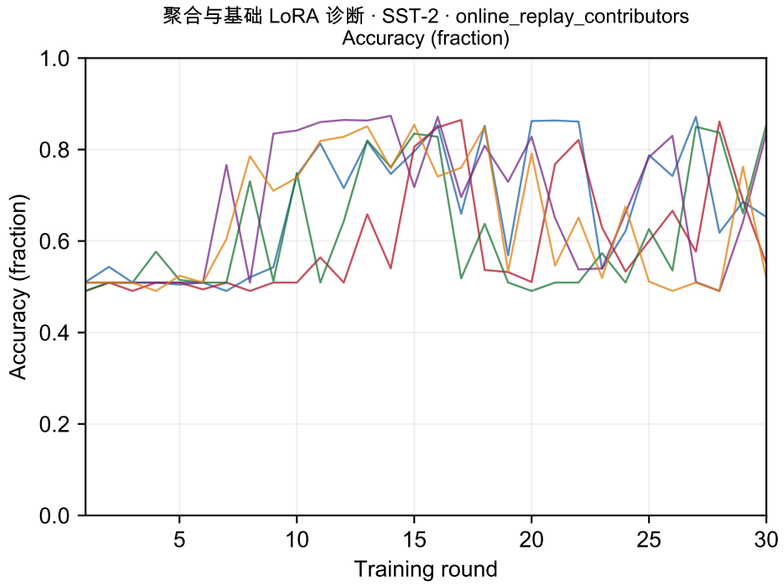


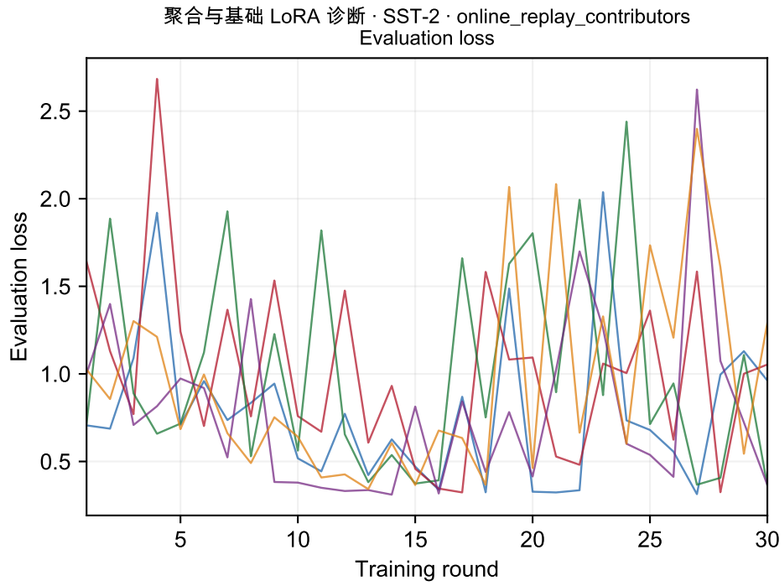


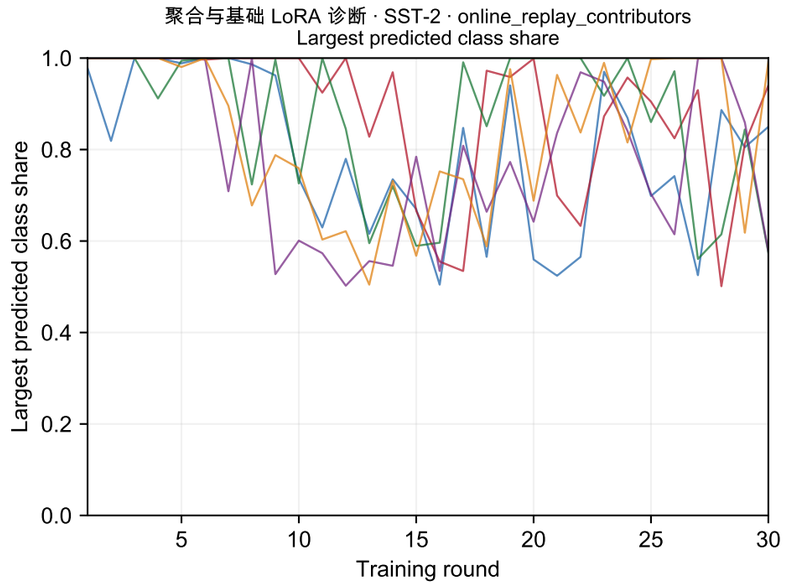


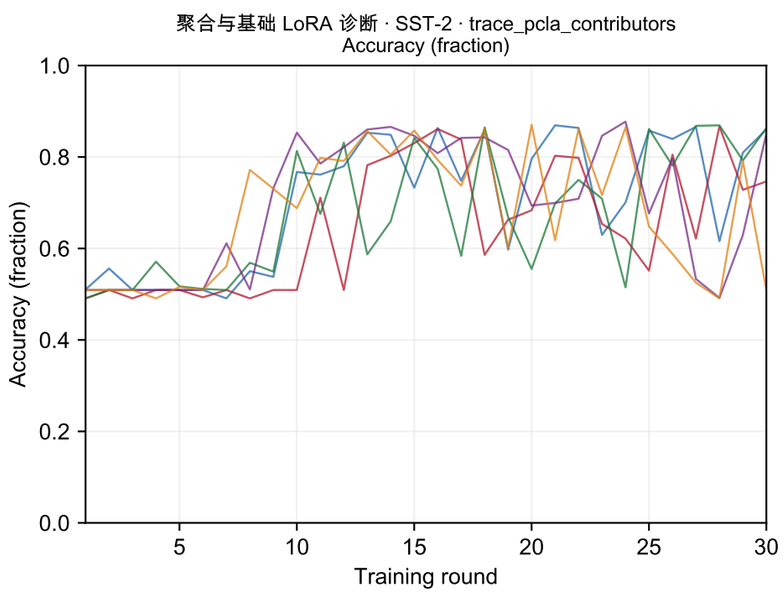


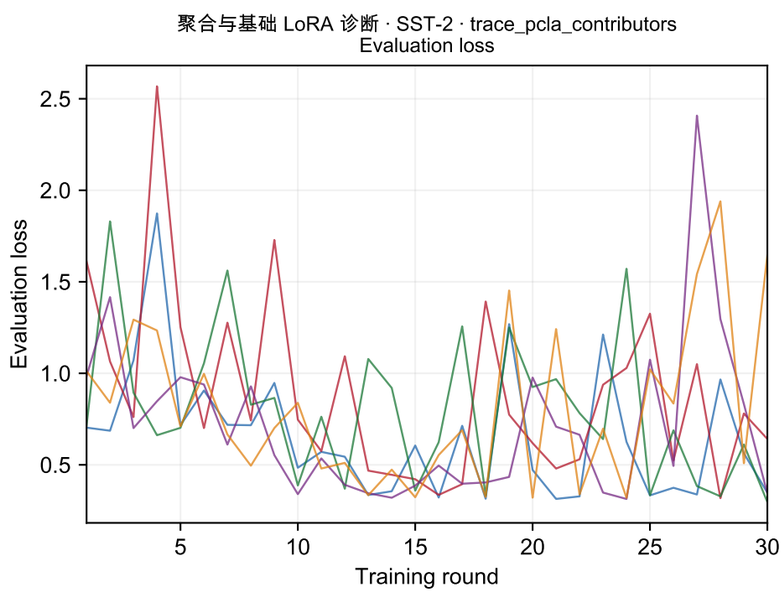
Full round trajectories · methods 9–10 of 45

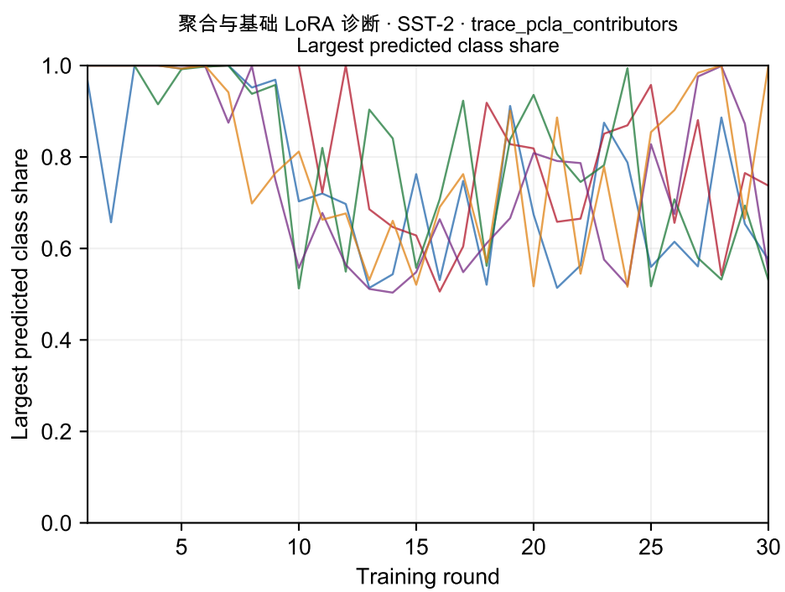


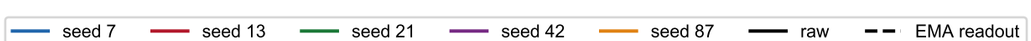






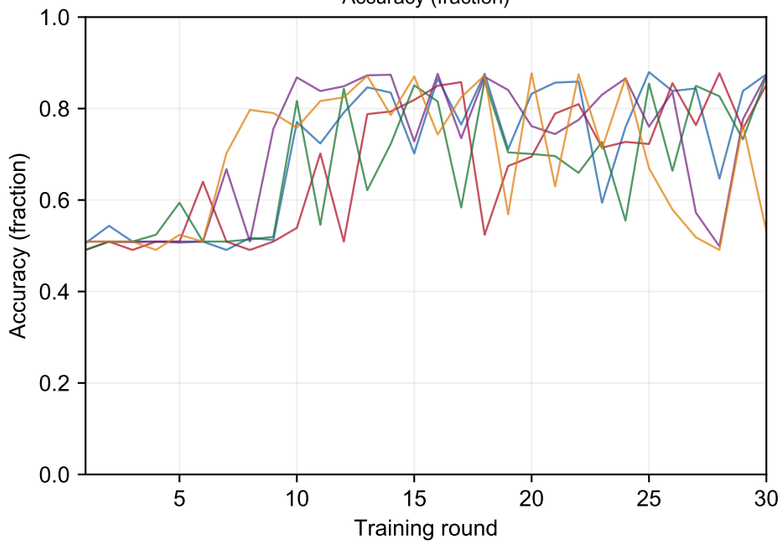




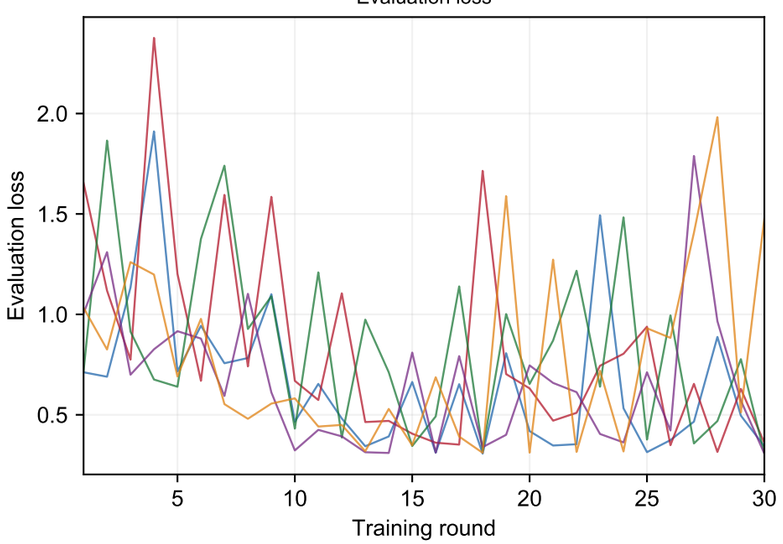


Full round trajectories · methods 11–12 of 45

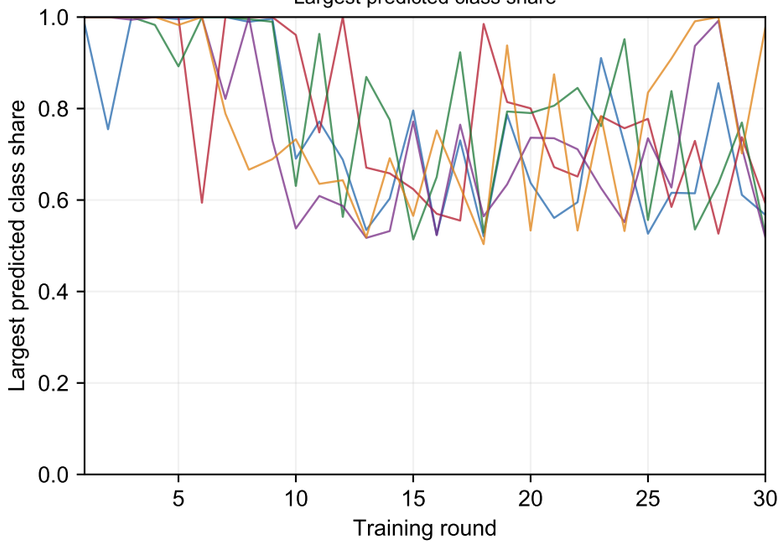
聚合与基础 LoRA 诊断 · SST-2 · trace_pcla_function_norm_matched
Accuracy (fraction)



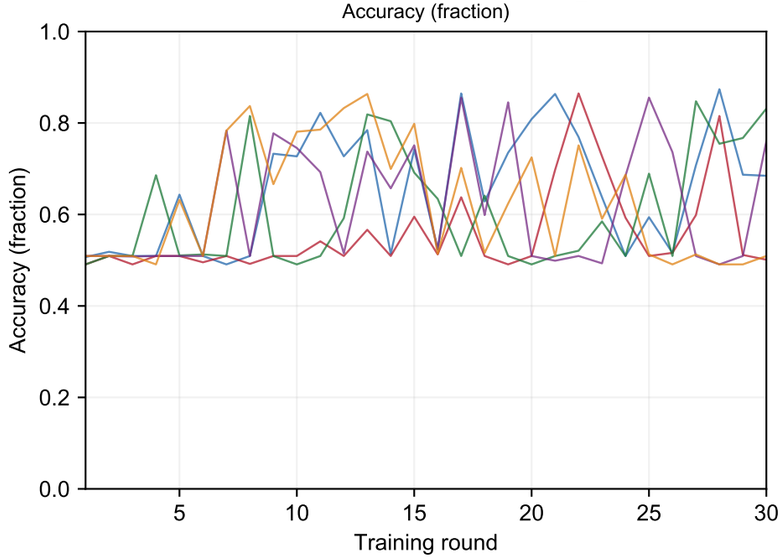
聚合与基础 LoRA 诊断 · SST-2 · trace_pcla_function_norm_matched
Evaluation loss



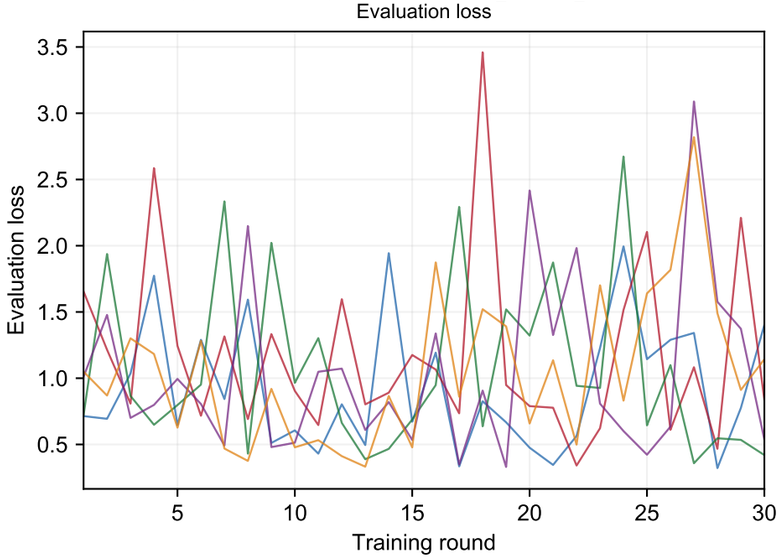
聚合与基础 LoRA 诊断 · SST-2 · trace_pcla_function_norm_matched
Largest predicted class share



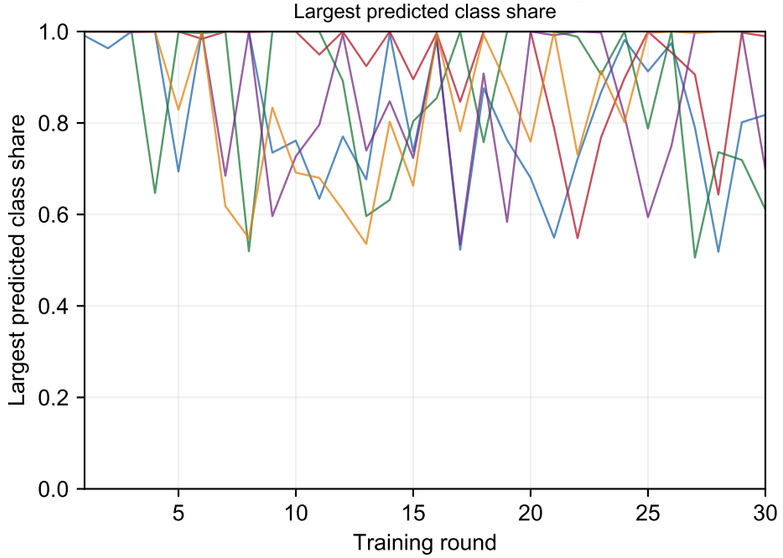
聚合与基础 LoRA 诊断 · SST-2 · trace_random_contributors
Accuracy (fraction)



聚合与基础 LoRA 诊断 · SST-2 · trace_random_contributors
Evaluation loss



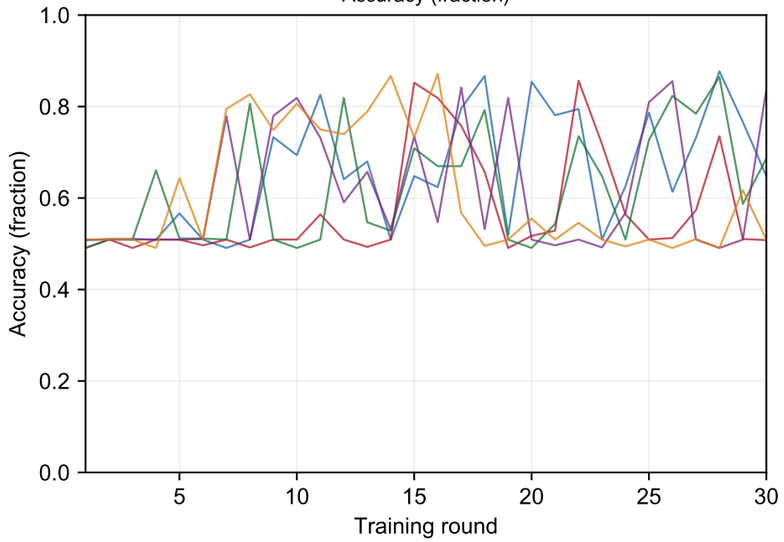
聚合与基础 LoRA 诊断 · SST-2 · trace_random_contributors
Largest predicted class share



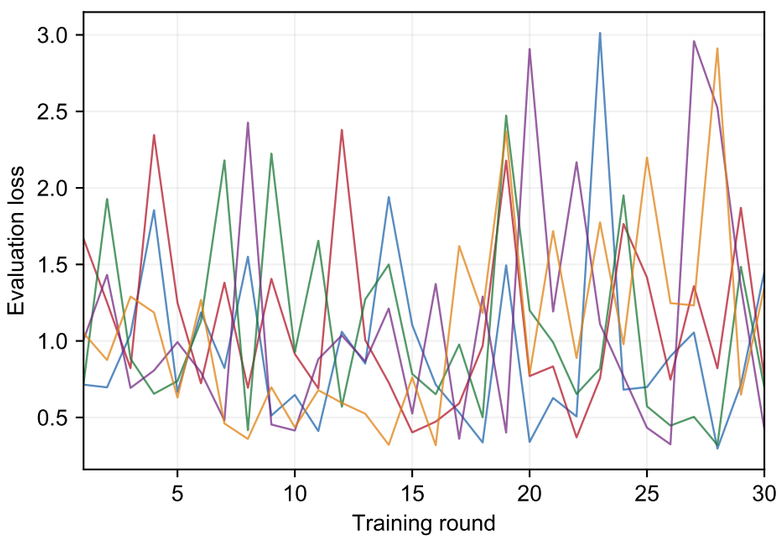


Full round trajectories · methods 13–14 of 45

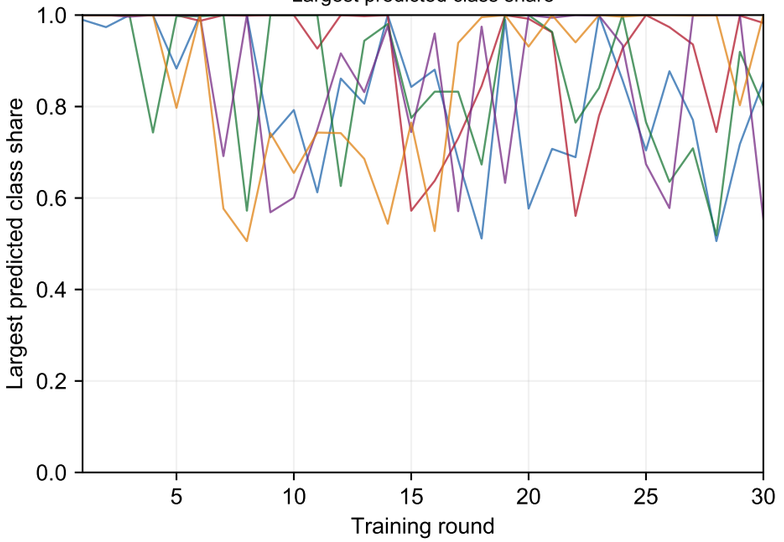
聚合与基础 LoRA 诊断 · SST-2 · trace_random_function_norm_matched
Accuracy (fraction)



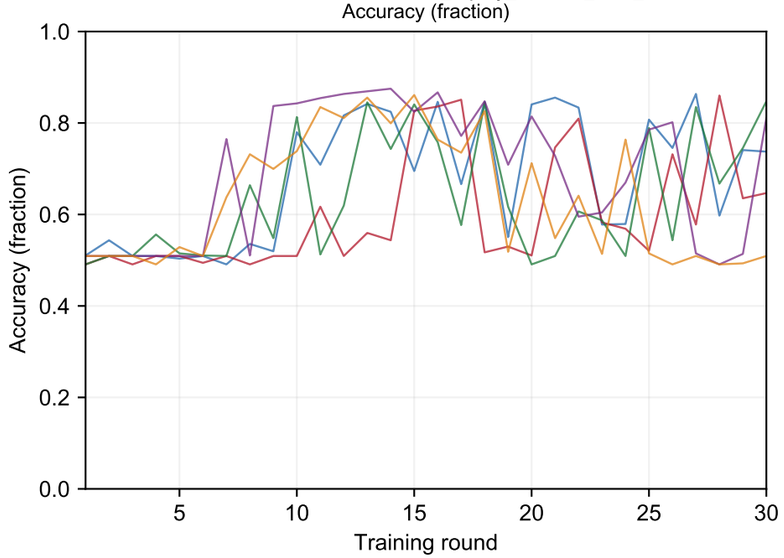
聚合与基础 LoRA 诊断 · SST-2 · trace_random_function_norm_matched
Evaluation loss



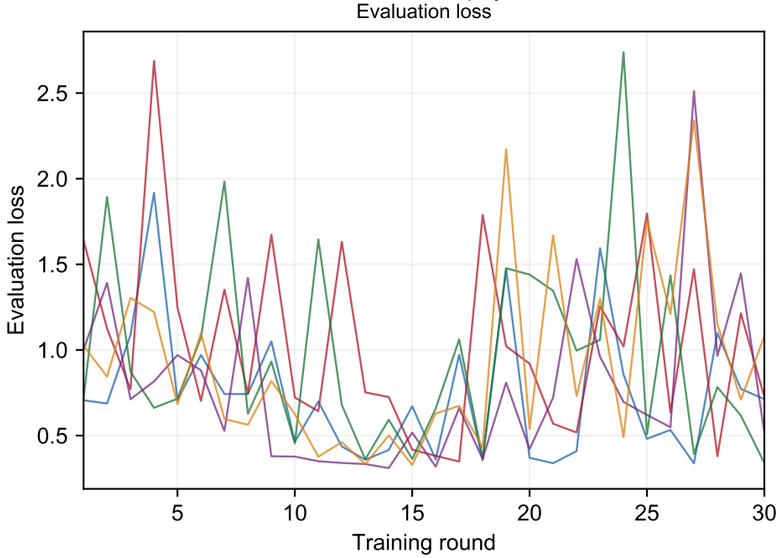
聚合与基础 LoRA 诊断 · SST-2 · trace_random_function_norm_matched
Largest predicted class share



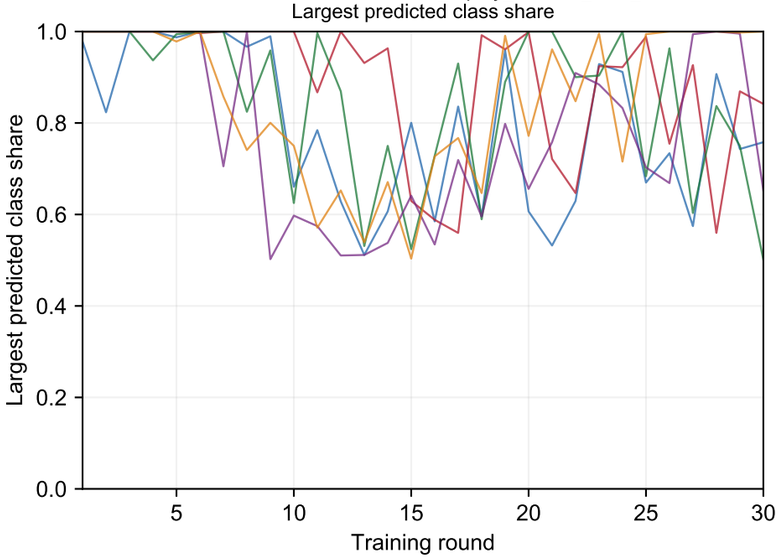
聚合与基础 LoRA 诊断 · SST-2 · trace_replay_random_rank_contributors
Accuracy (fraction)



聚合与基础 LoRA 诊断 · SST-2 · trace_replay_random_rank_contributors
Evaluation loss

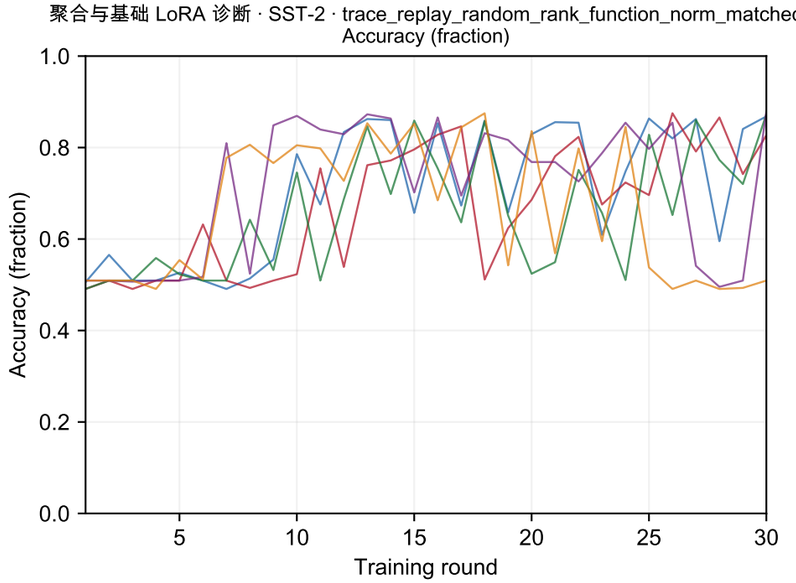


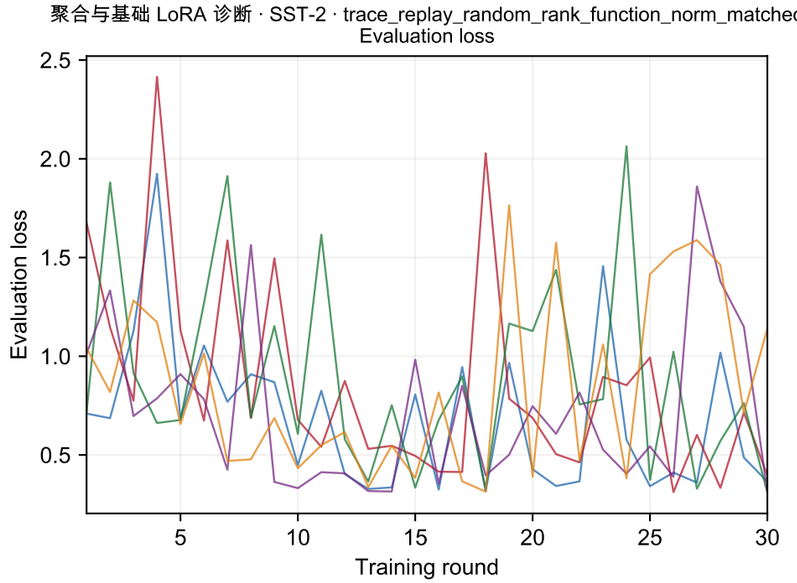
聚合与基础 LoRA 诊断 · SST-2 · trace_replay_random_rank_contributors
Largest predicted class share

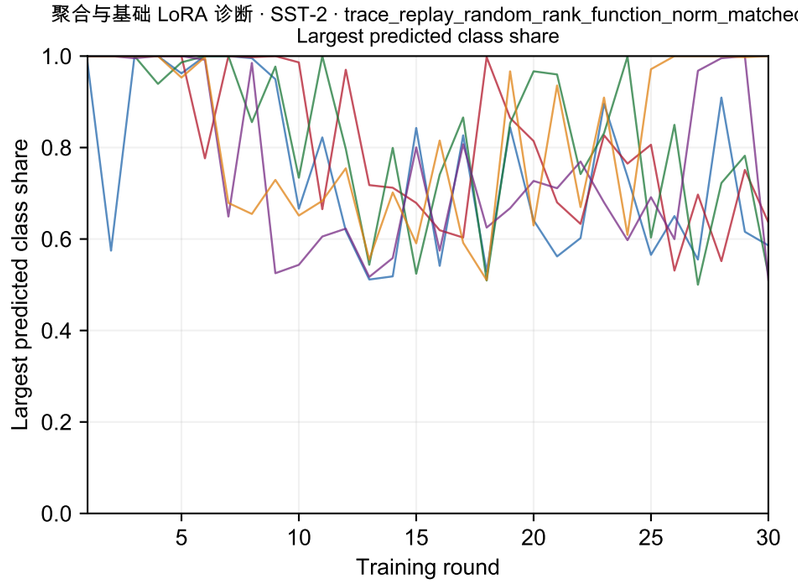


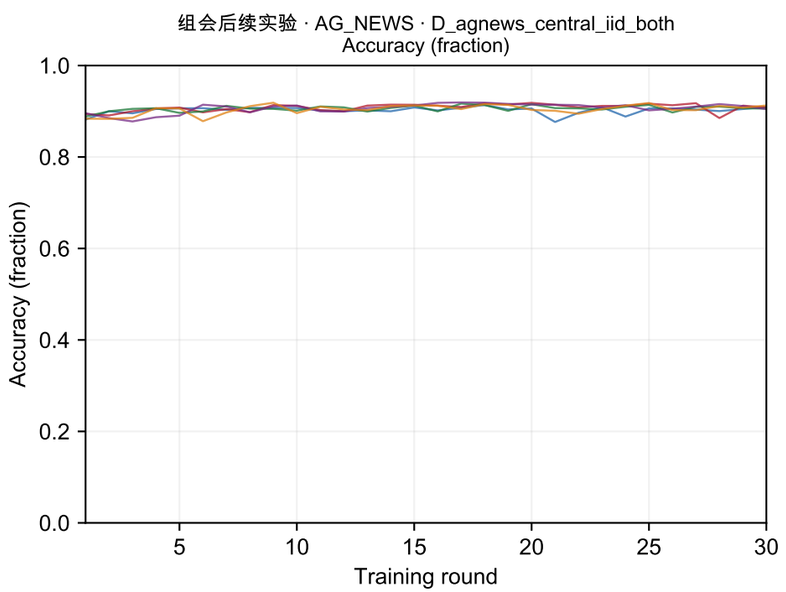


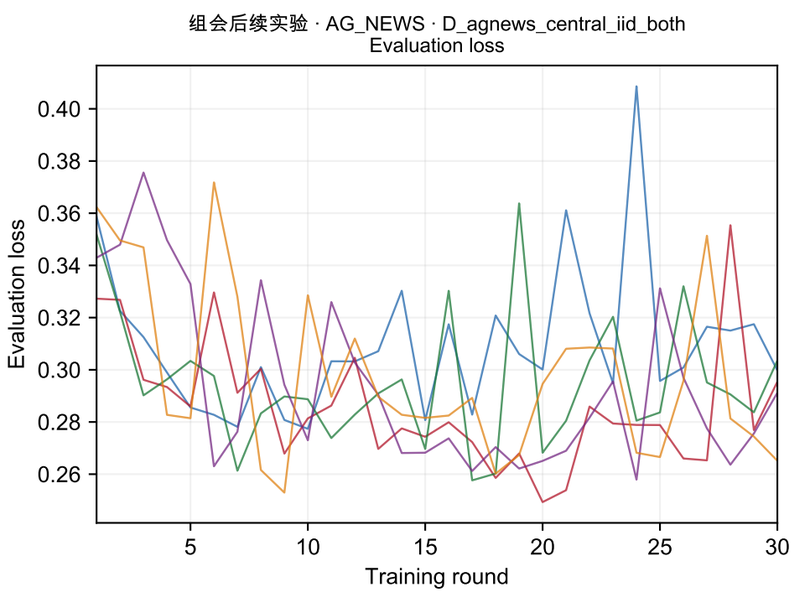
Full round trajectories · methods 15–16 of 45

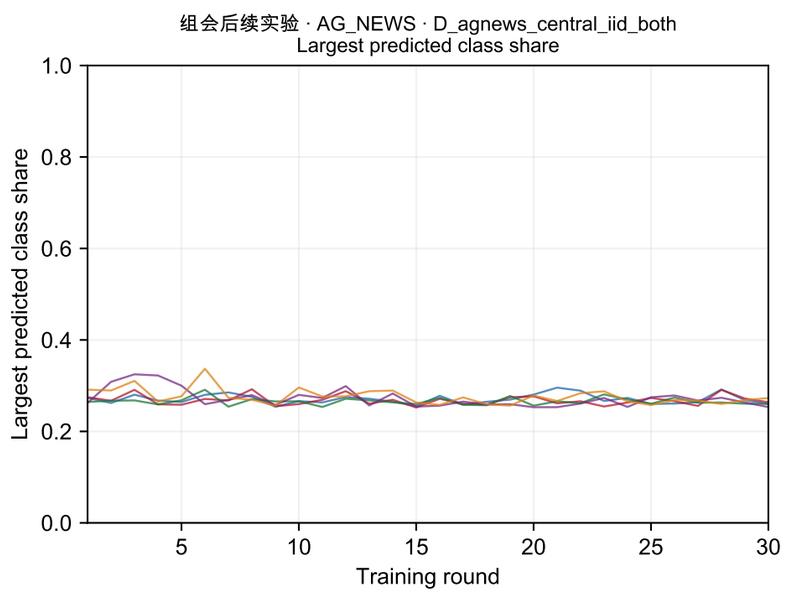


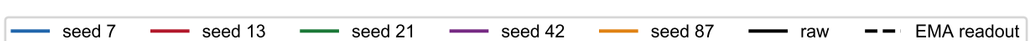




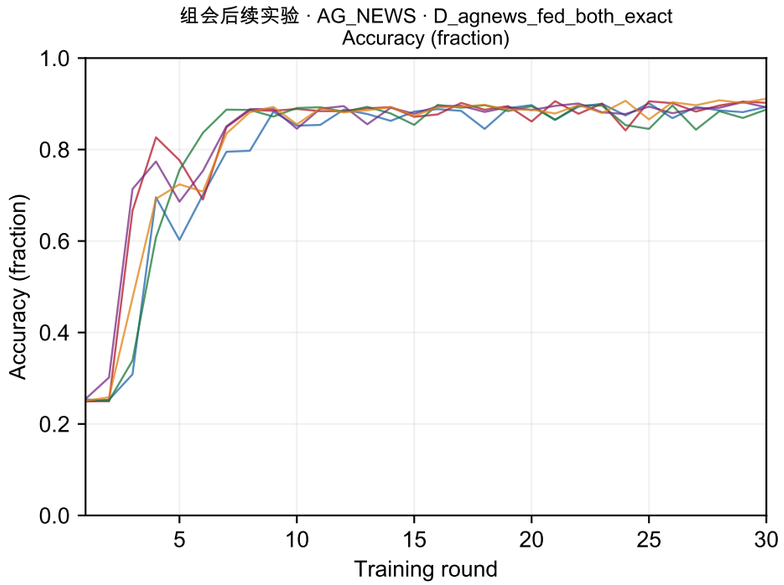


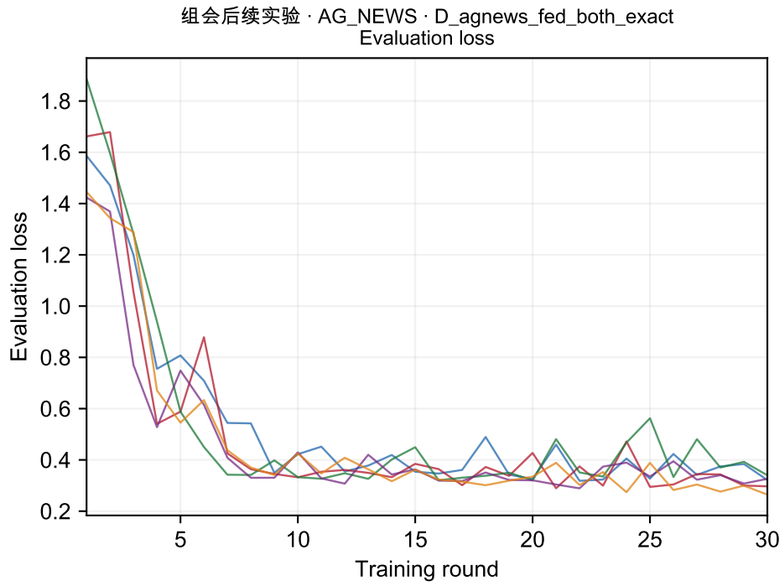


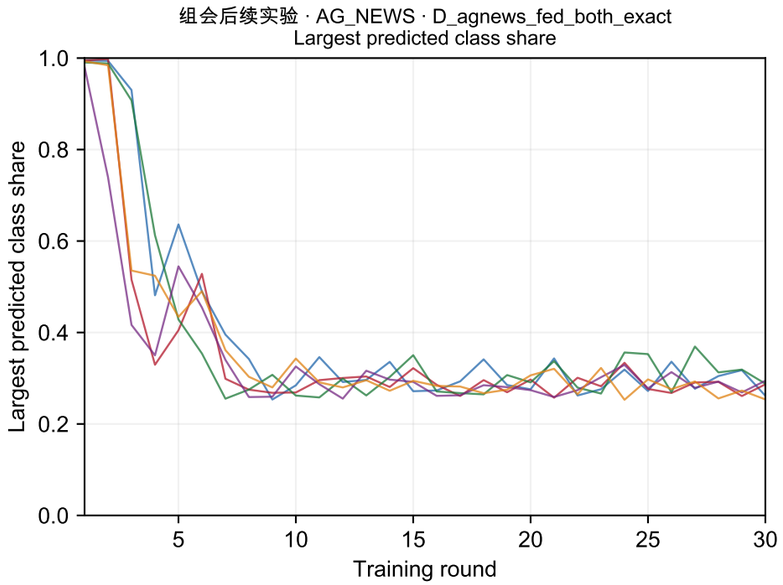


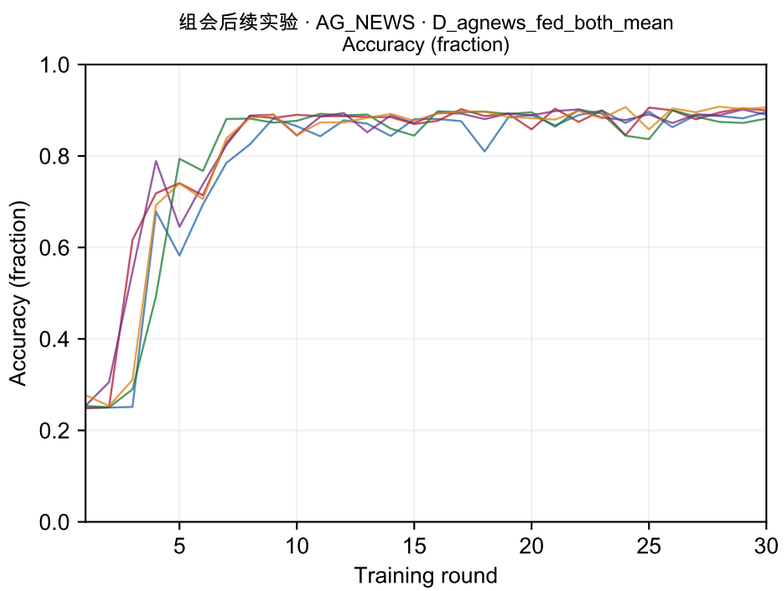


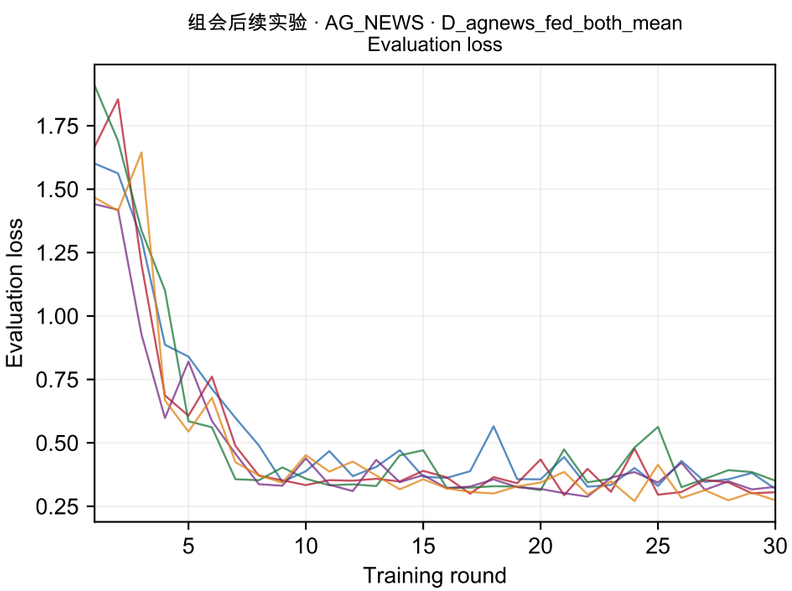
Full round trajectories · methods 17–18 of 45

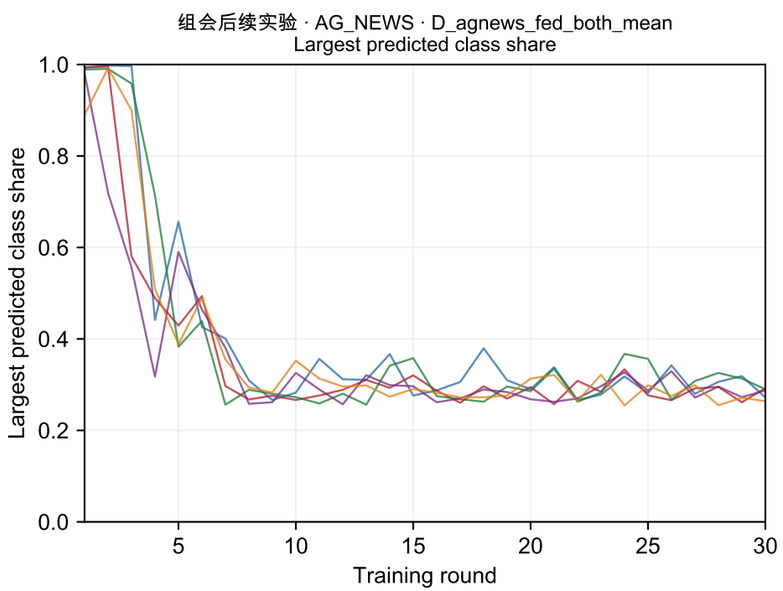


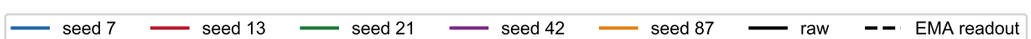




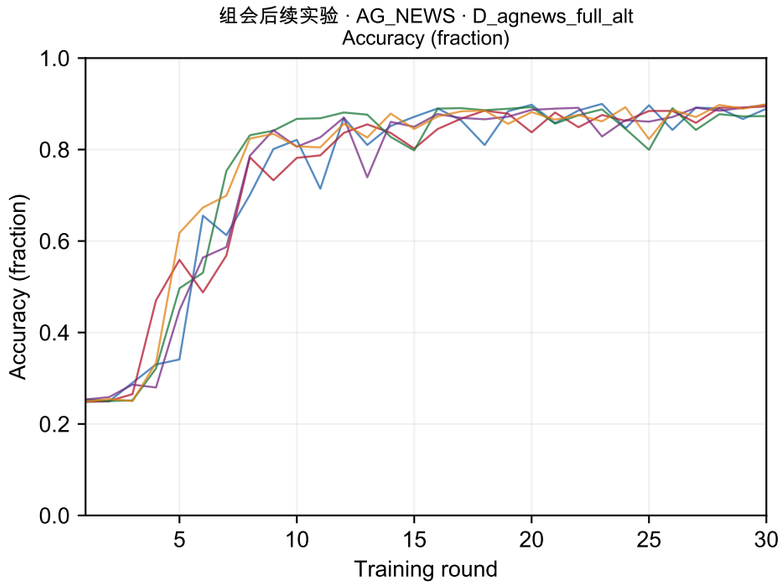


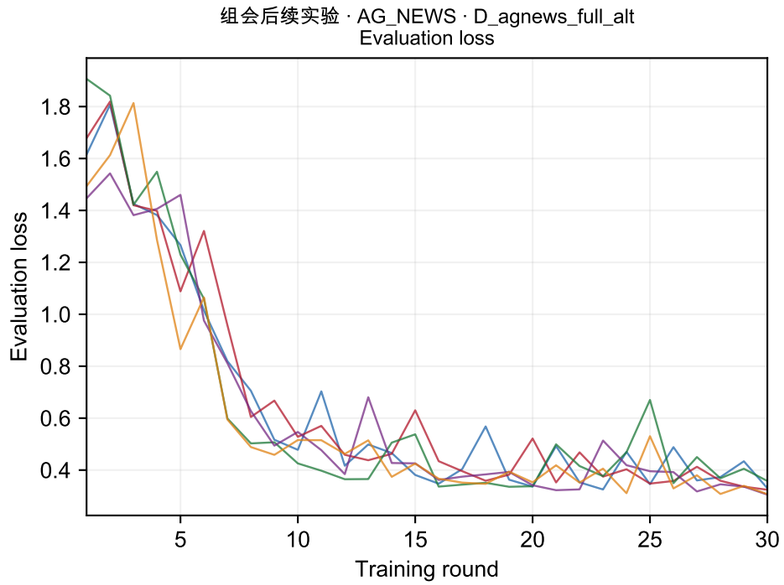


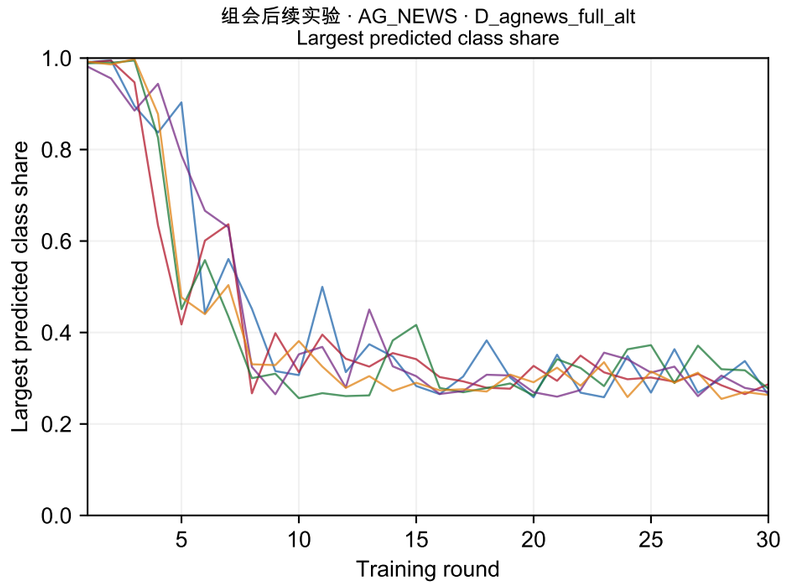


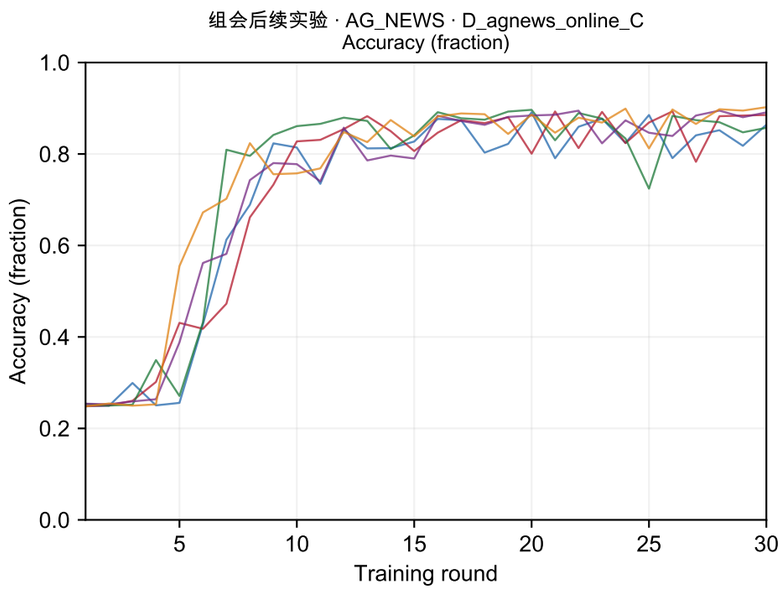


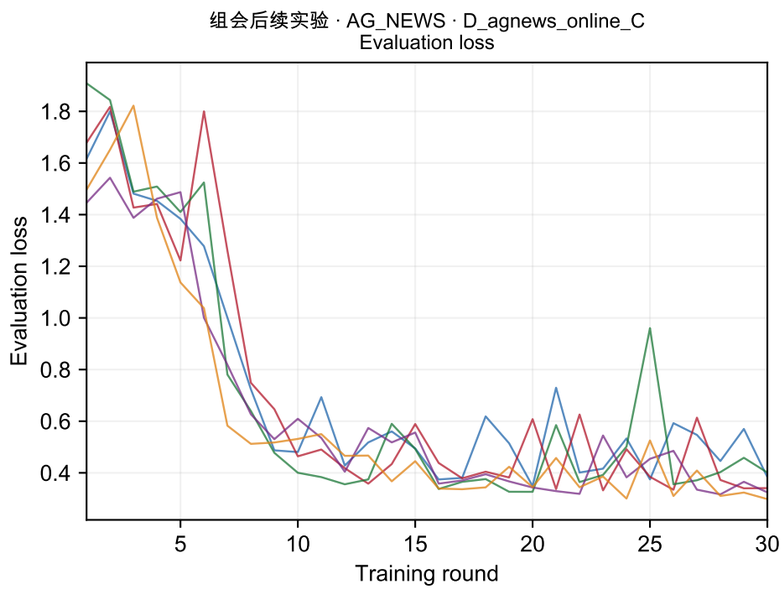
Full round trajectories · methods 19–20 of 45

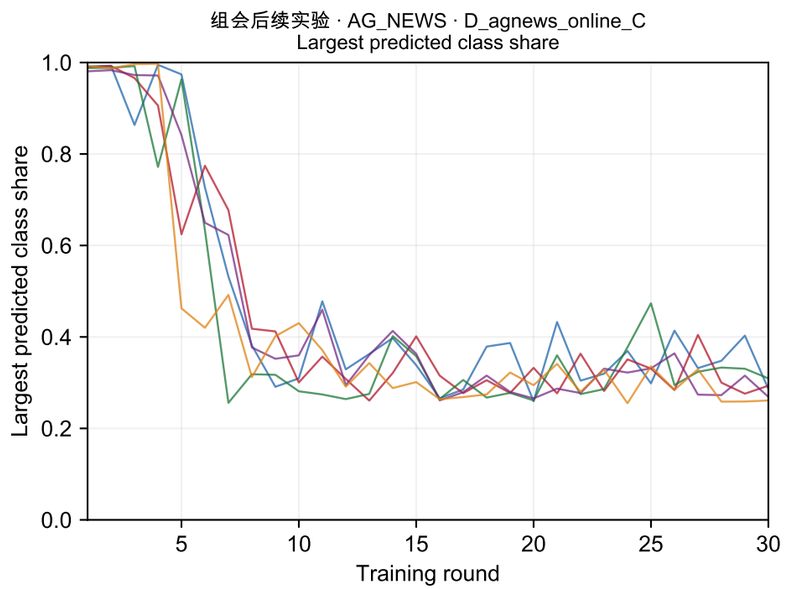






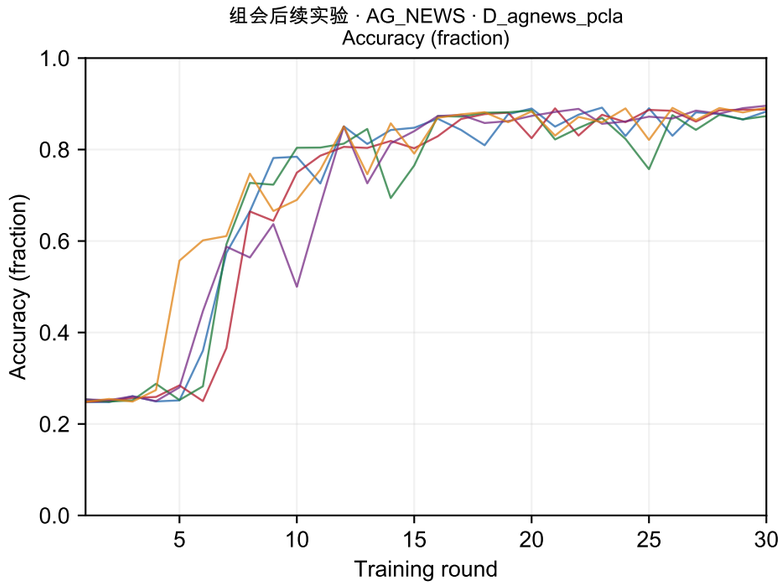


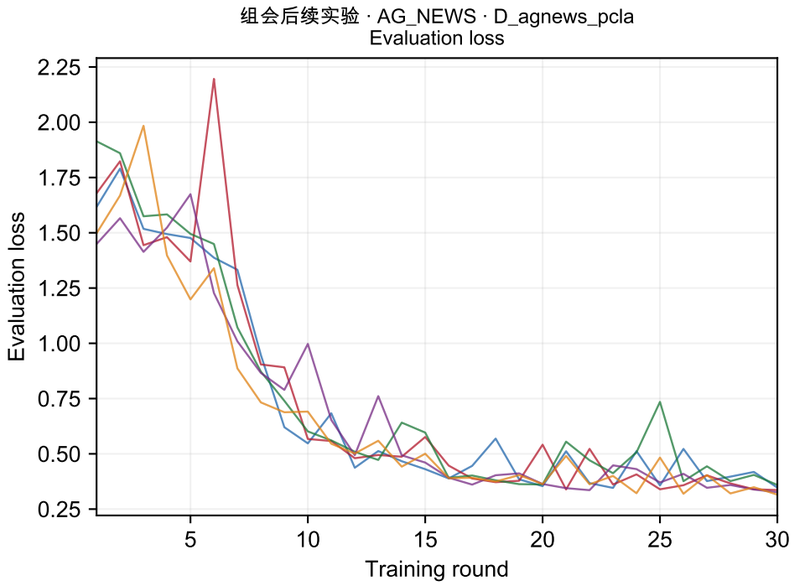


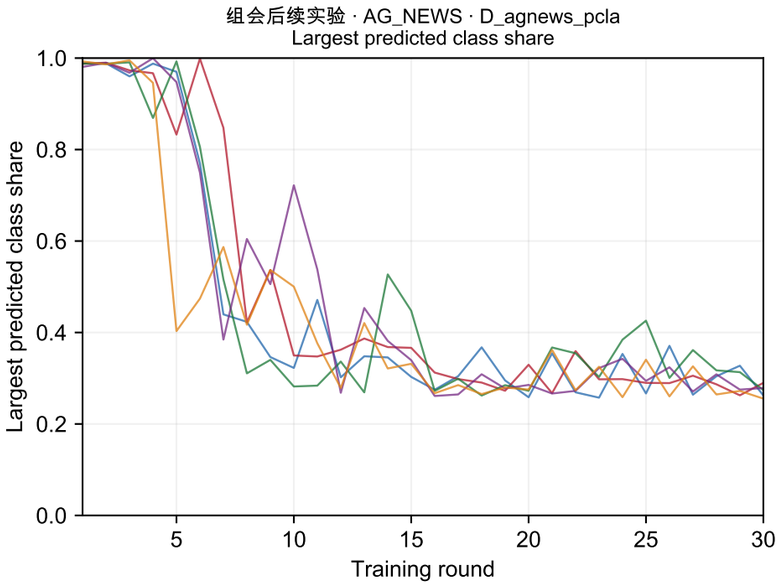


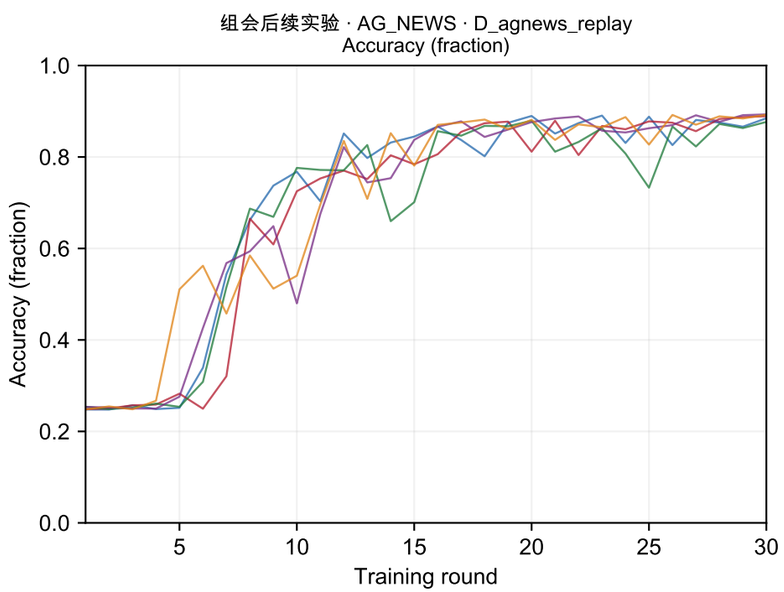


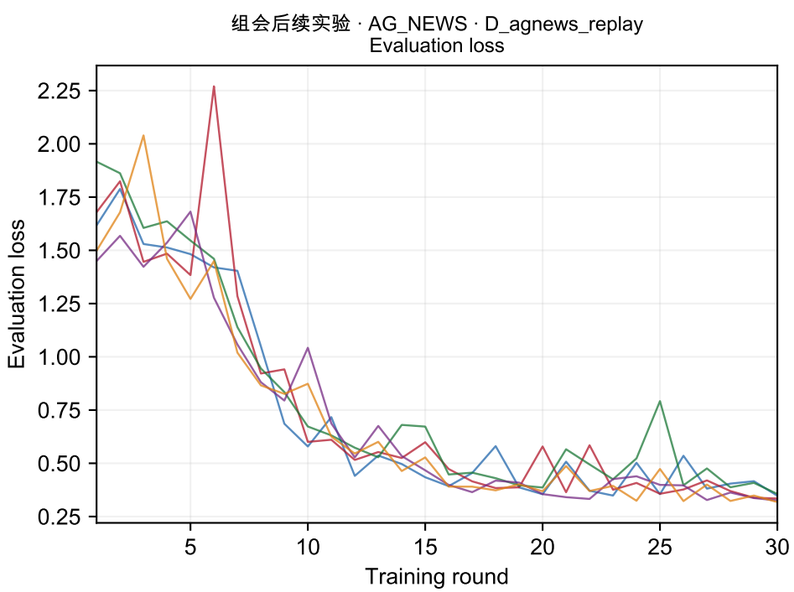
Full round trajectories · methods 21–22 of 45

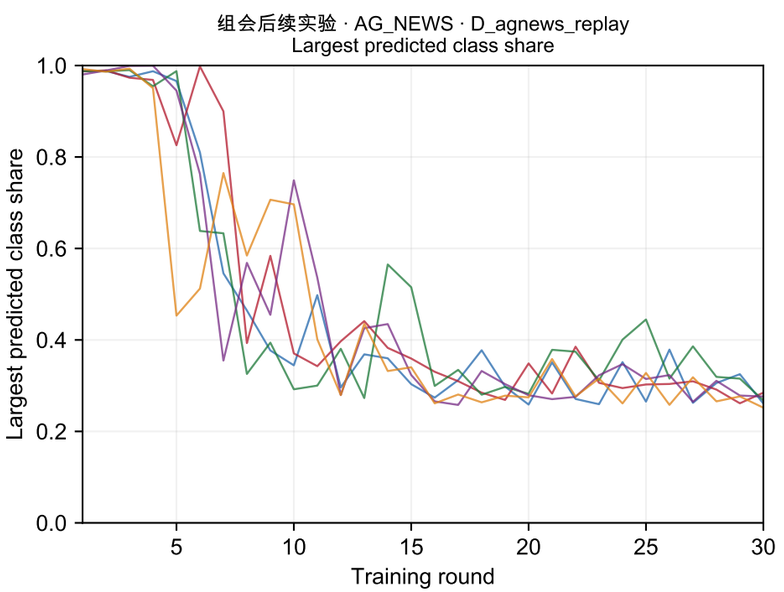


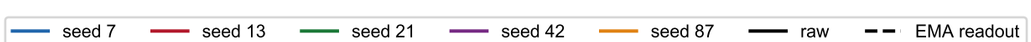




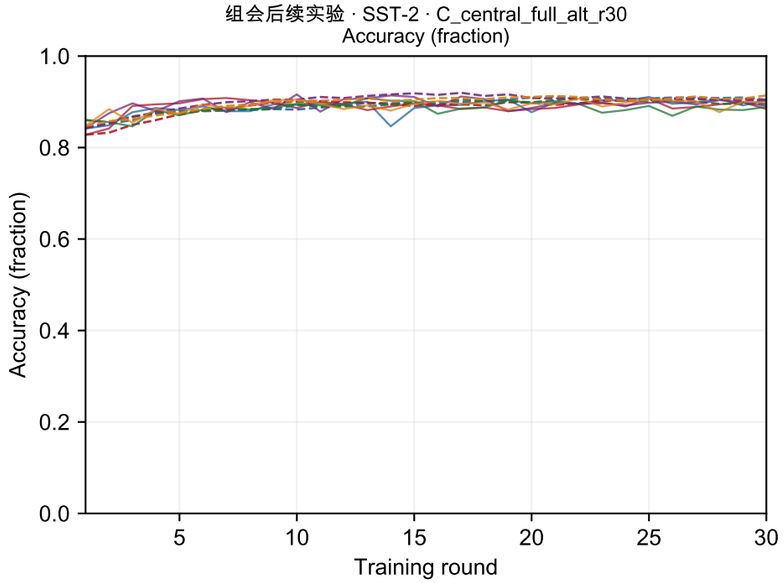


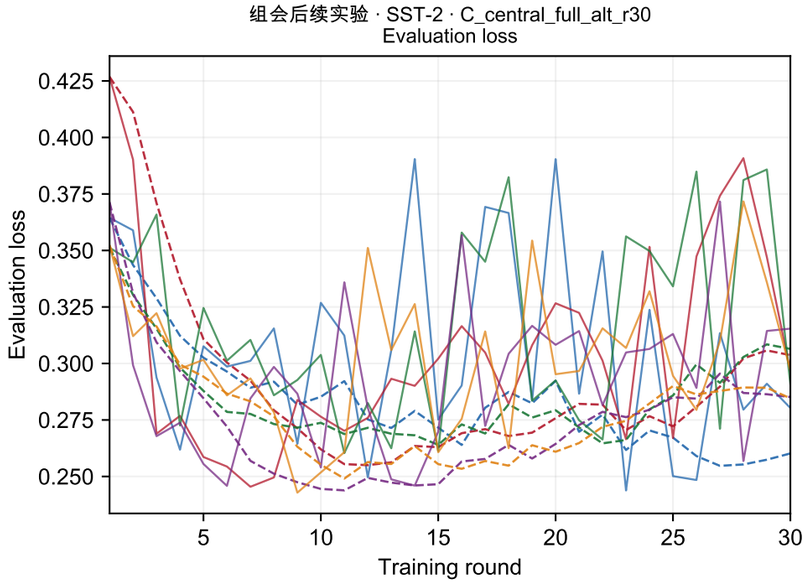


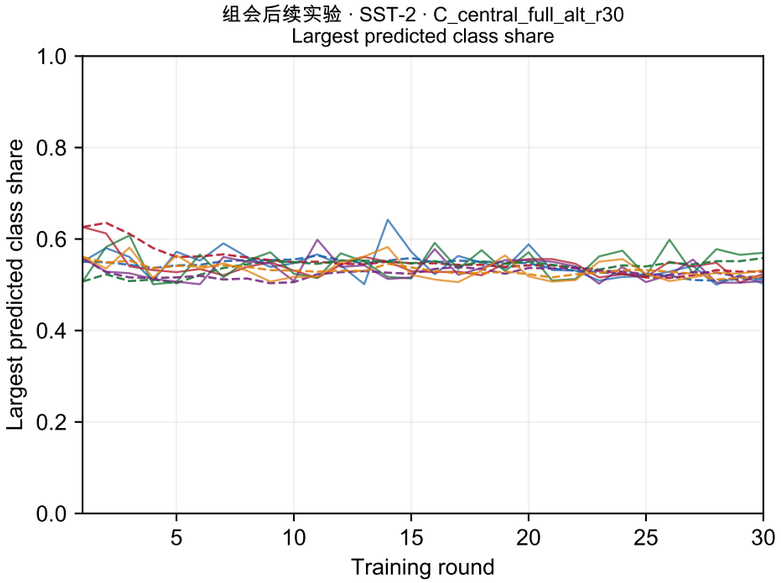


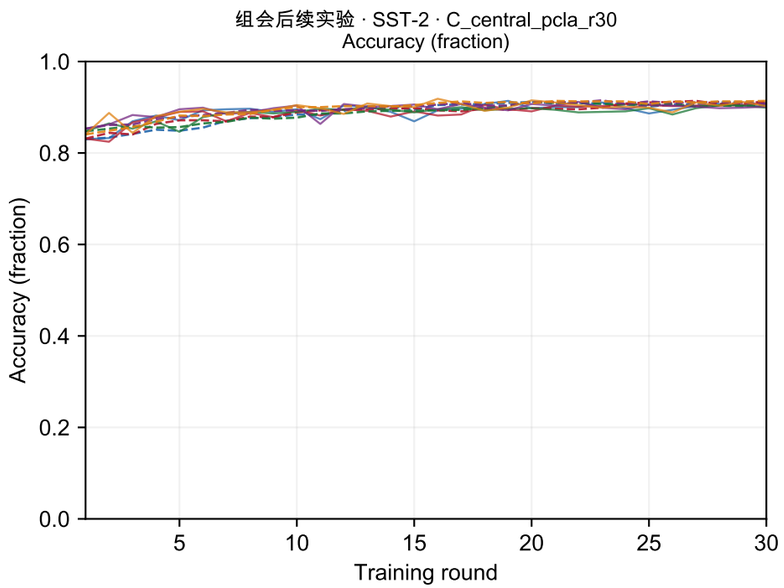


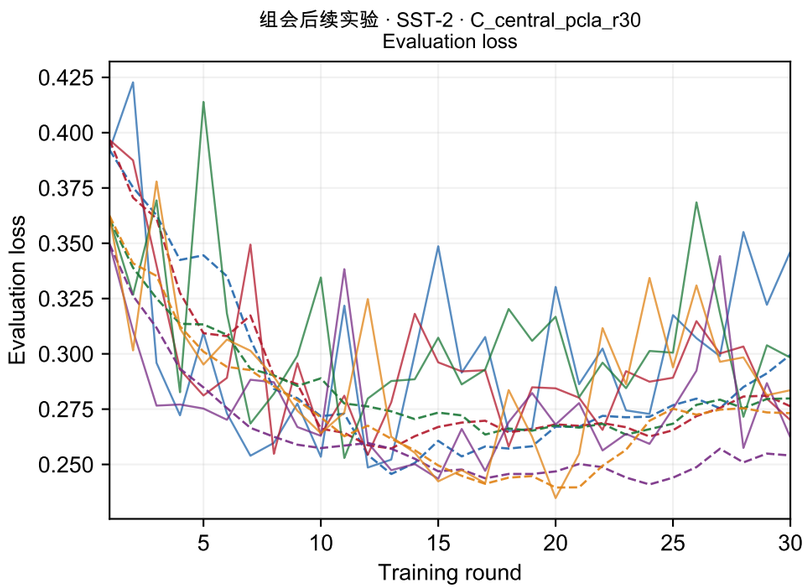
Full round trajectories · methods 23–24 of 45

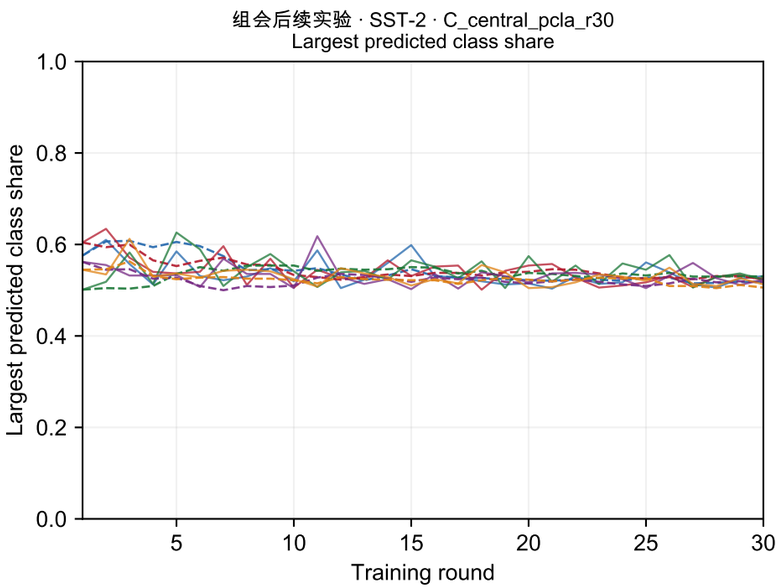






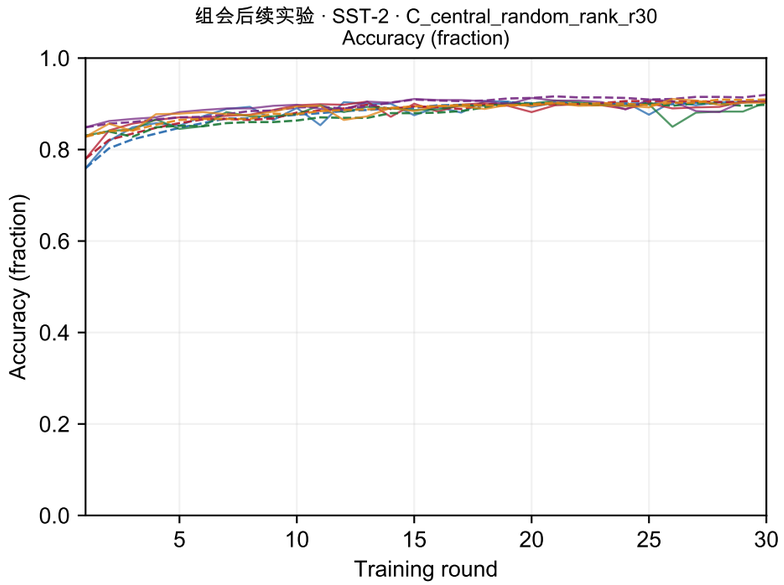


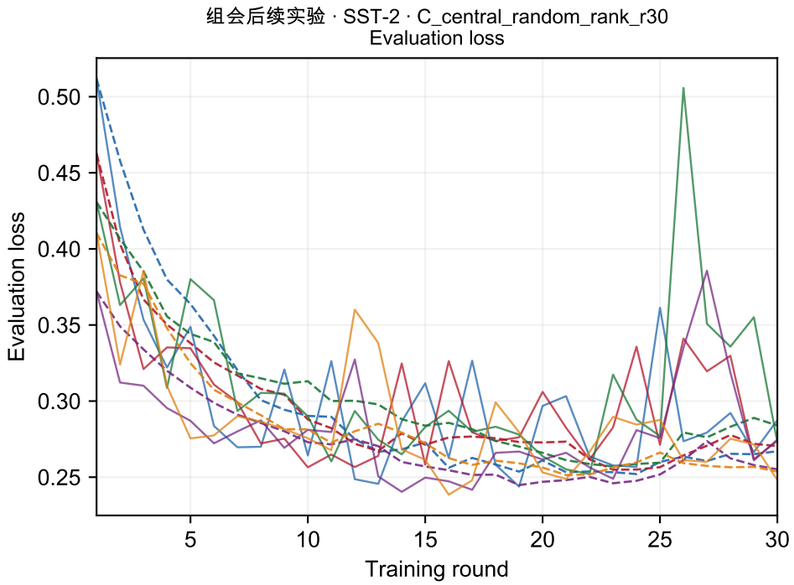


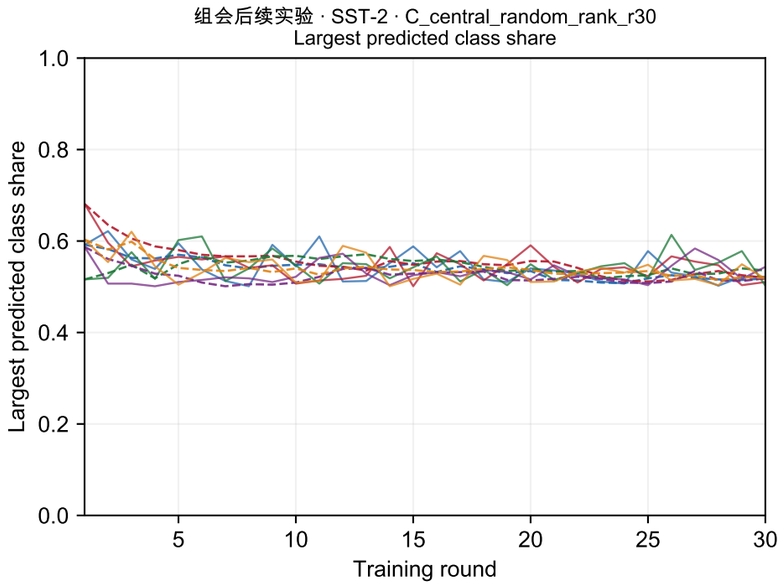


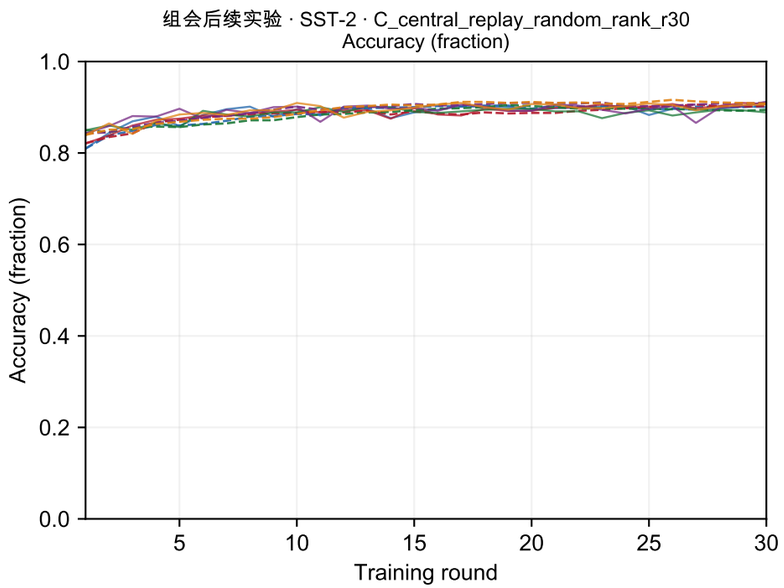


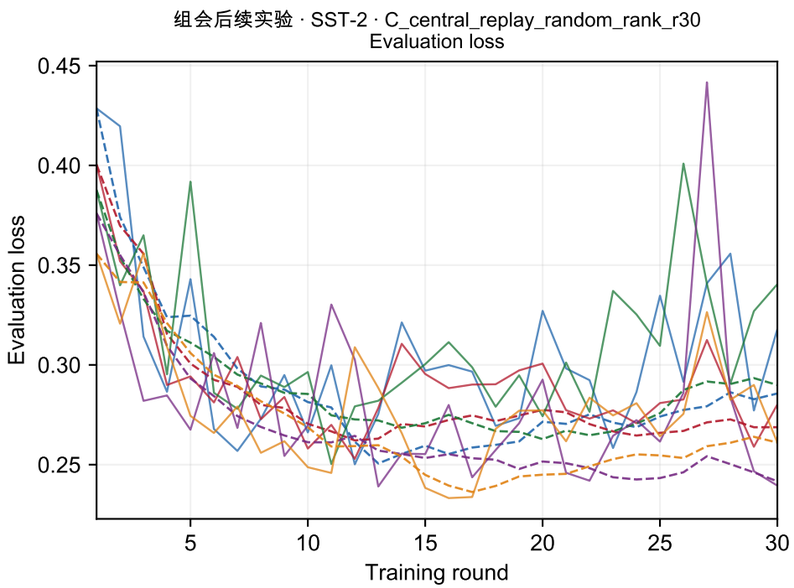
Full round trajectories · methods 25–26 of 45

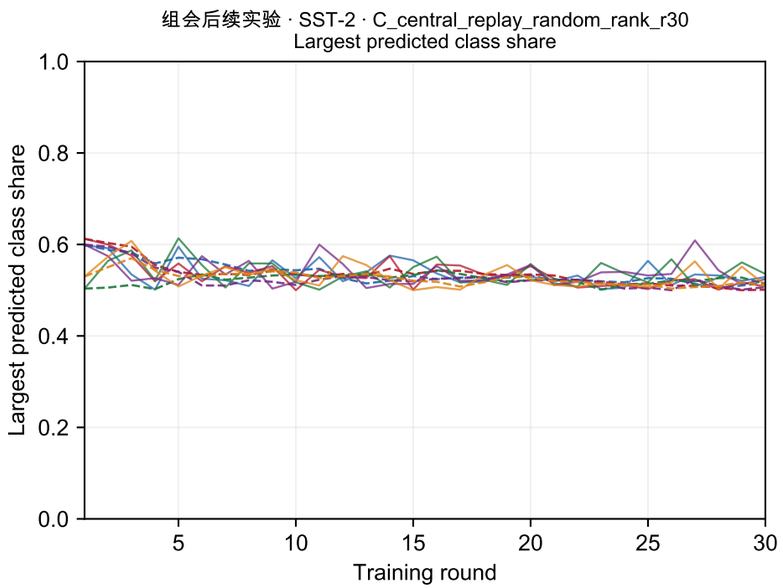






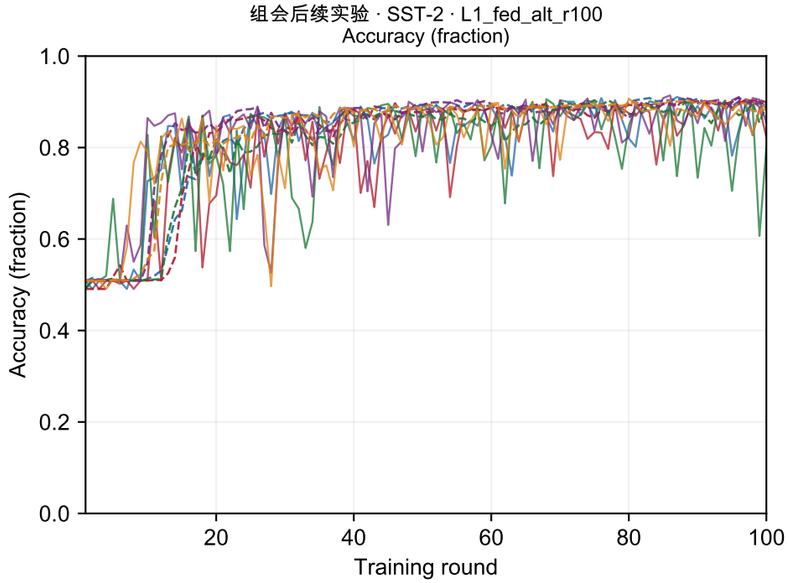


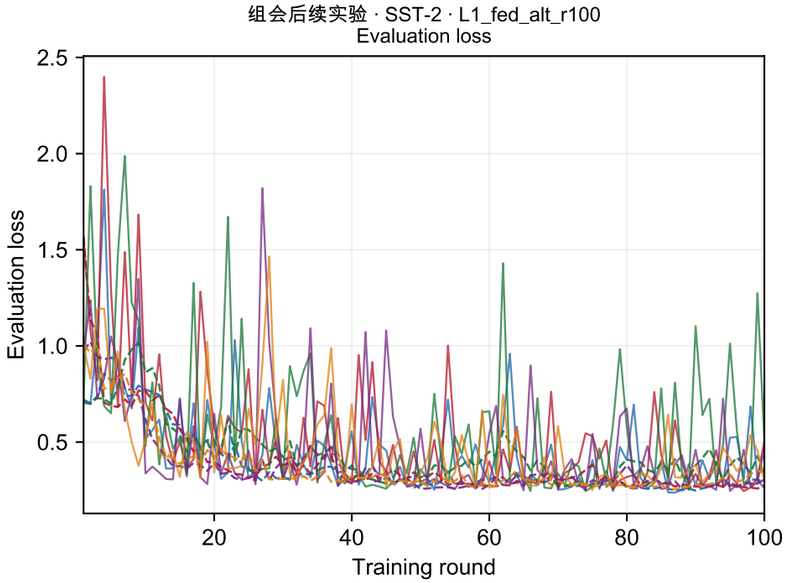


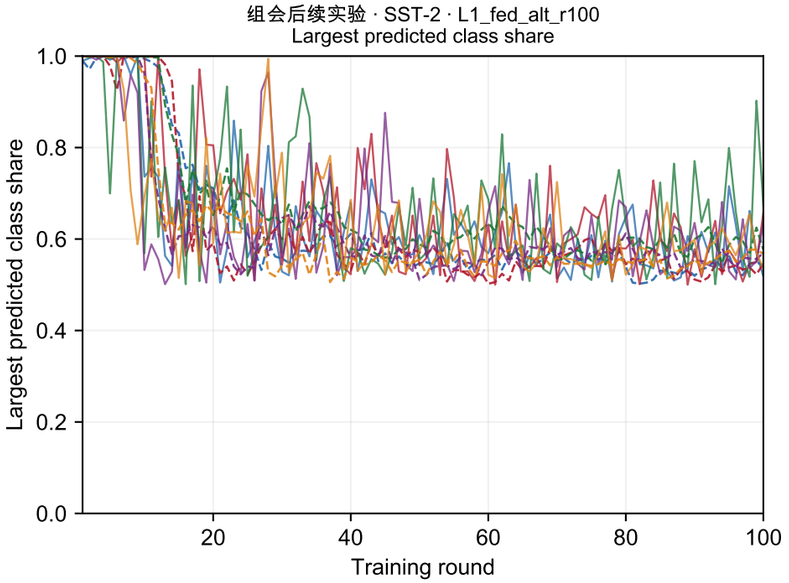


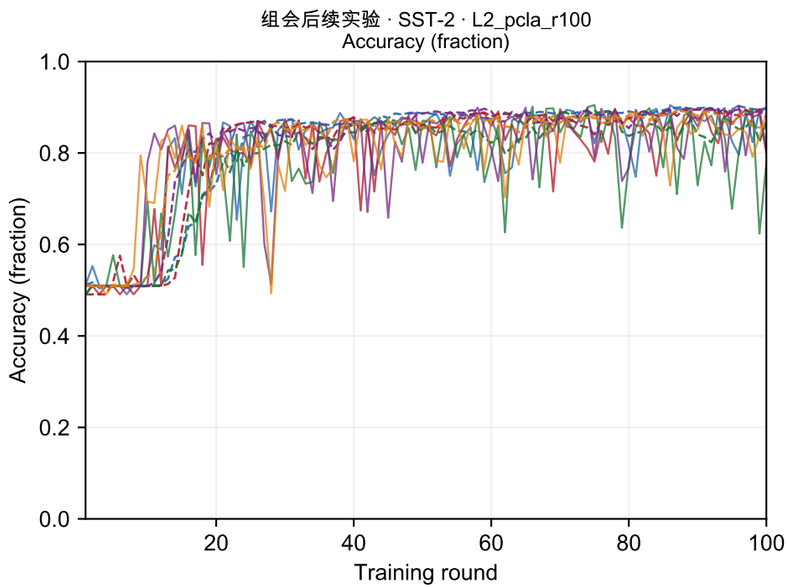


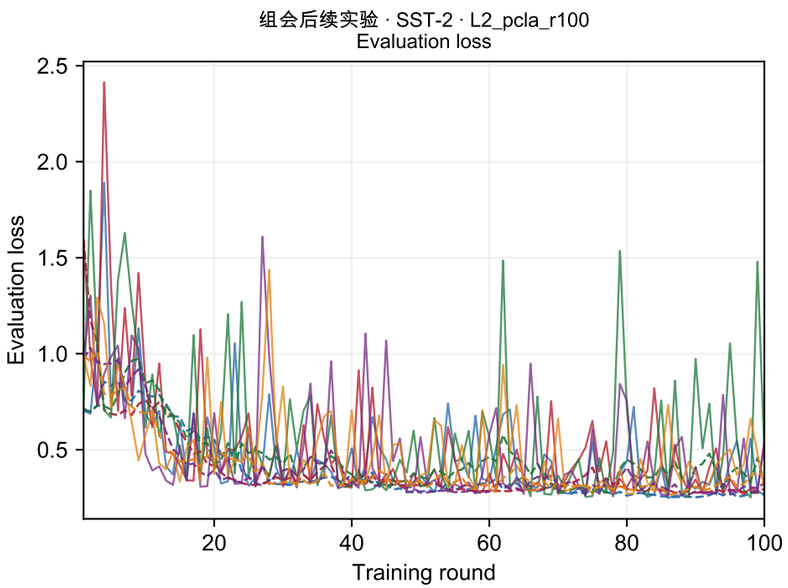
Full round trajectories · methods 27–28 of 45

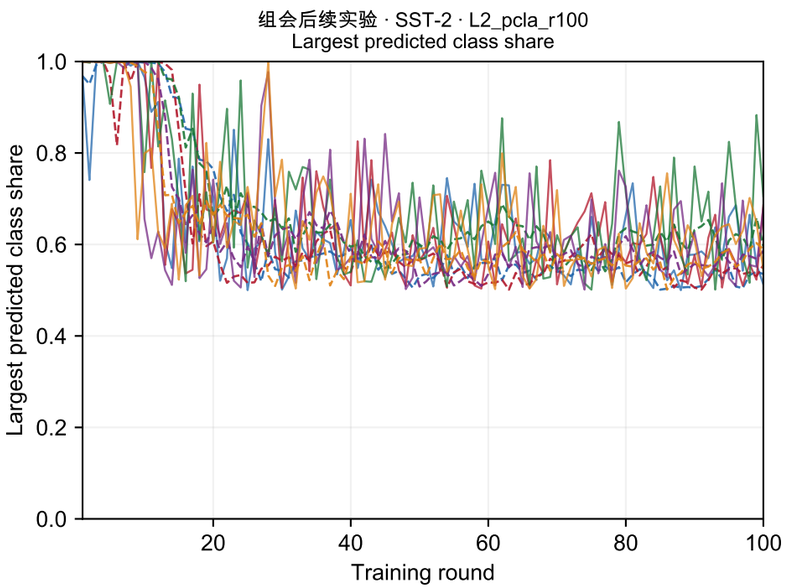






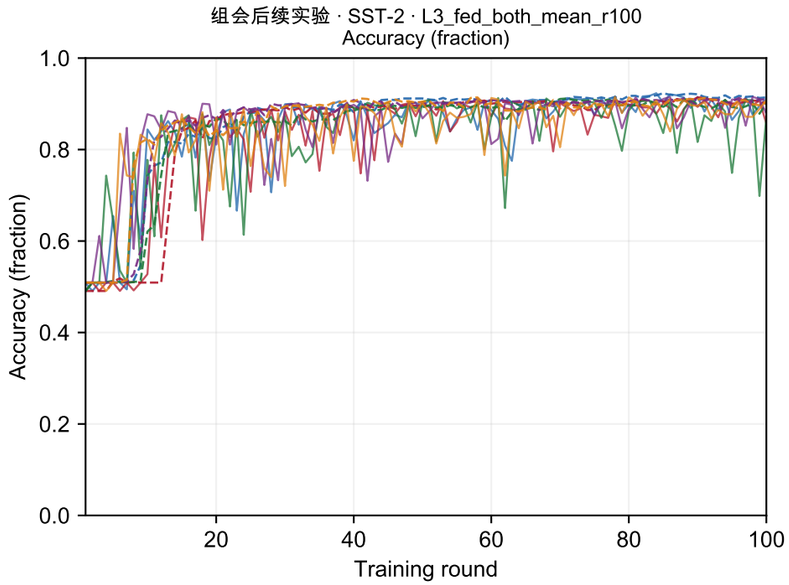


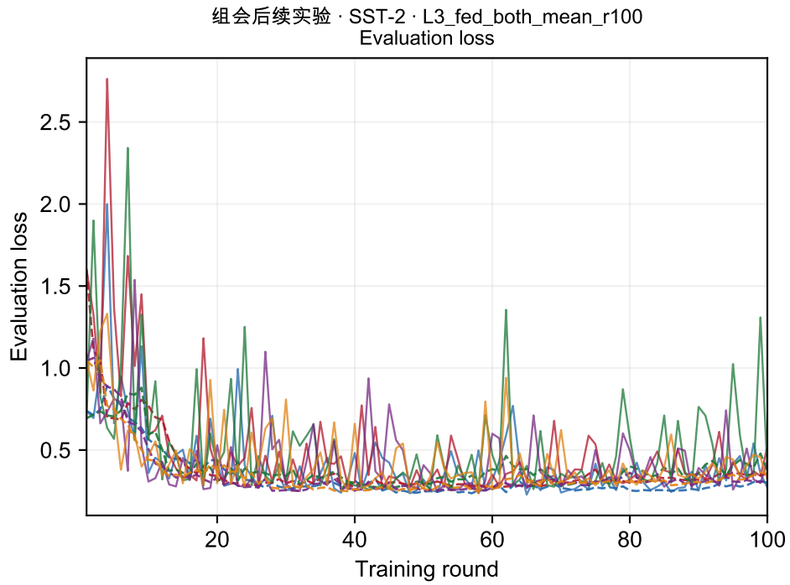


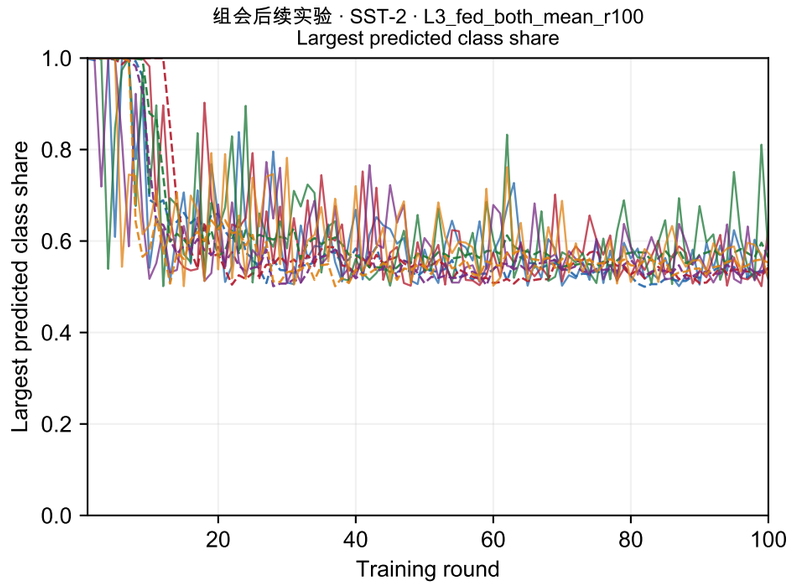


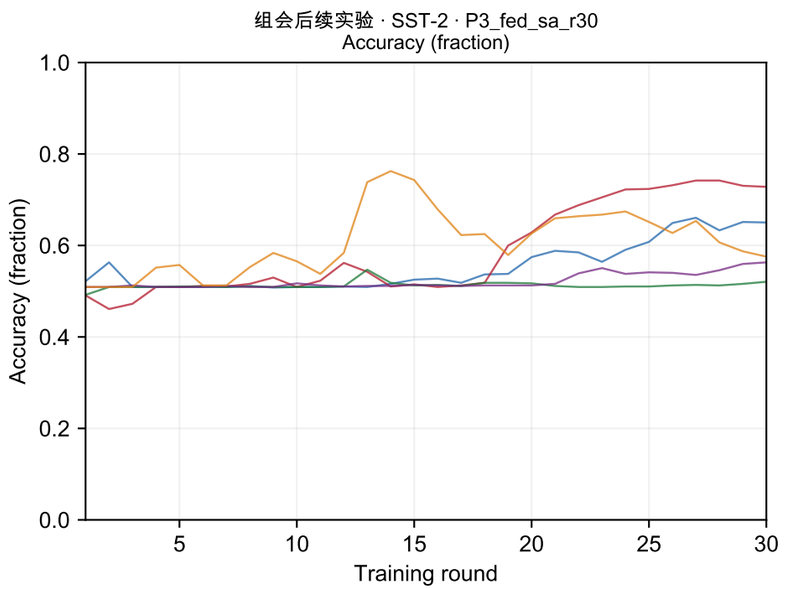


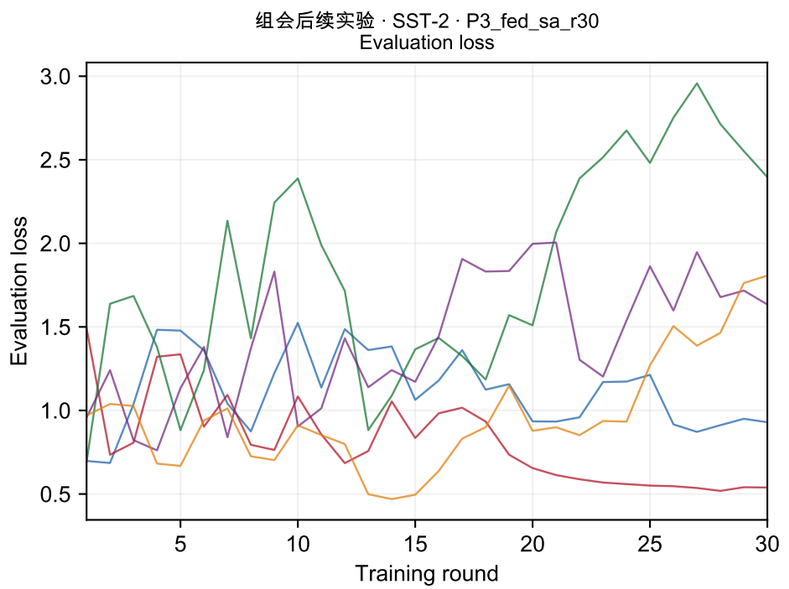
Full round trajectories · methods 29–30 of 45

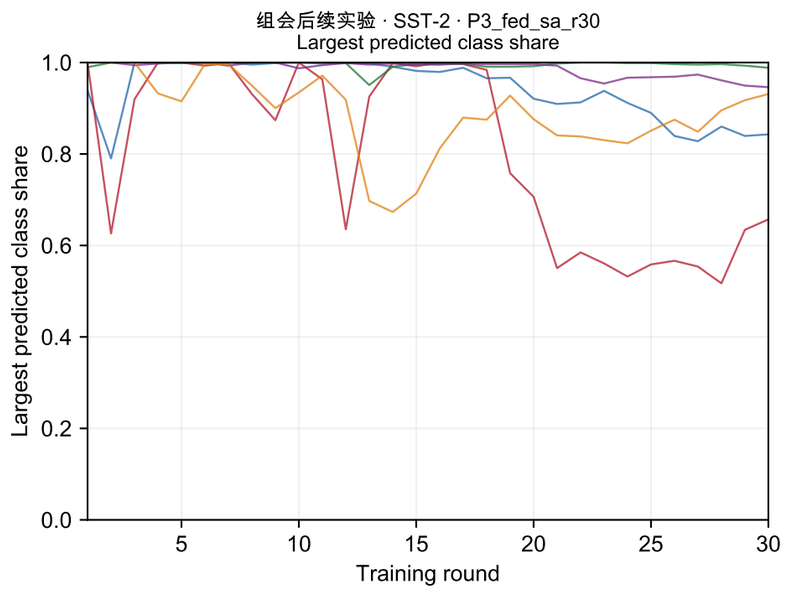






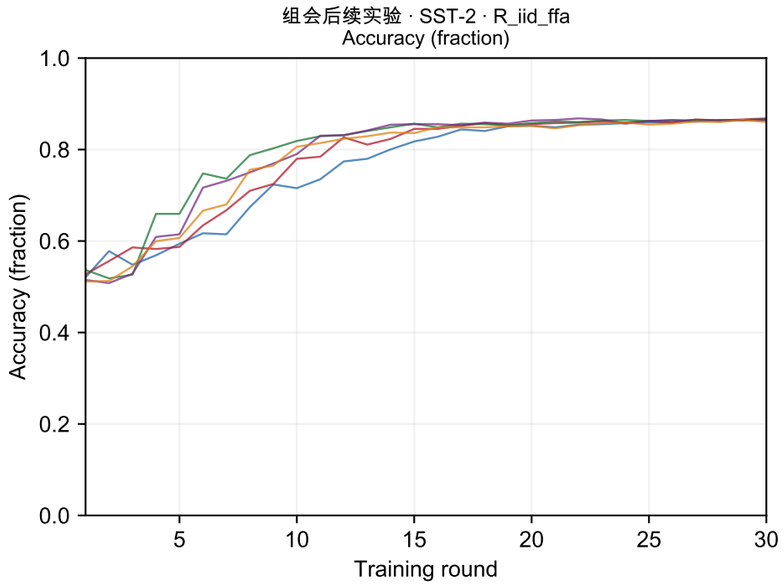


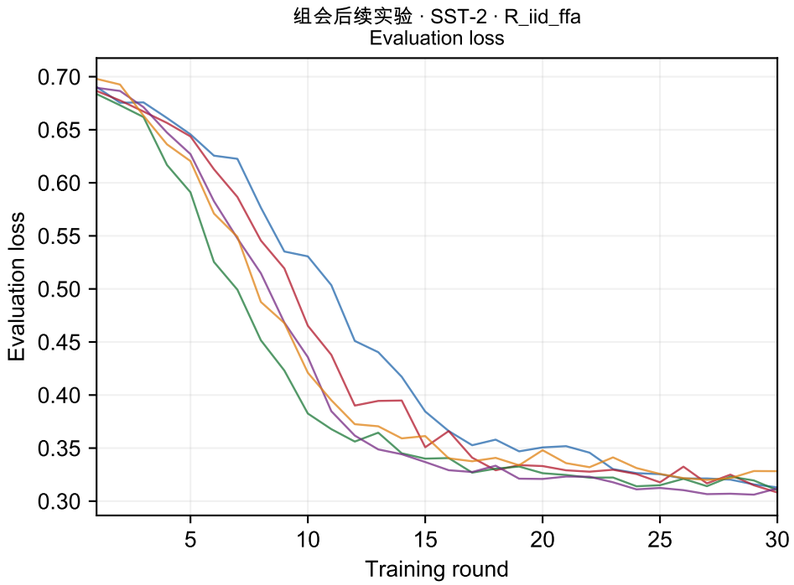


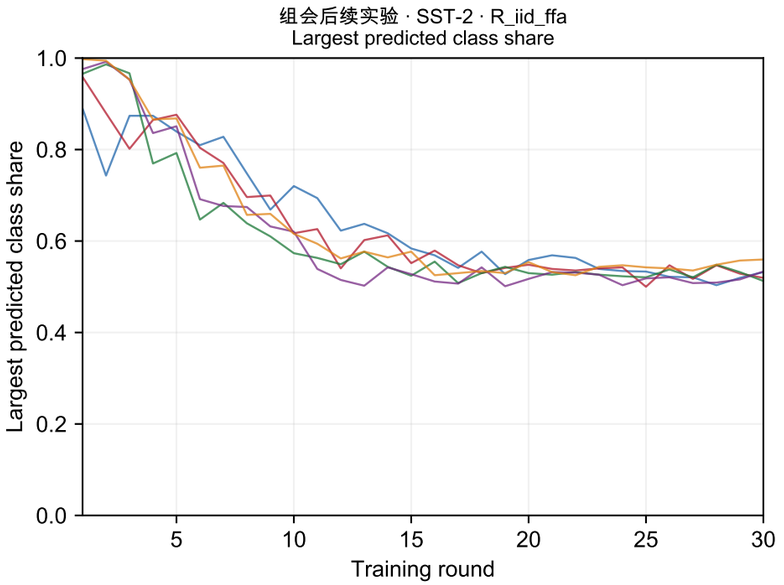


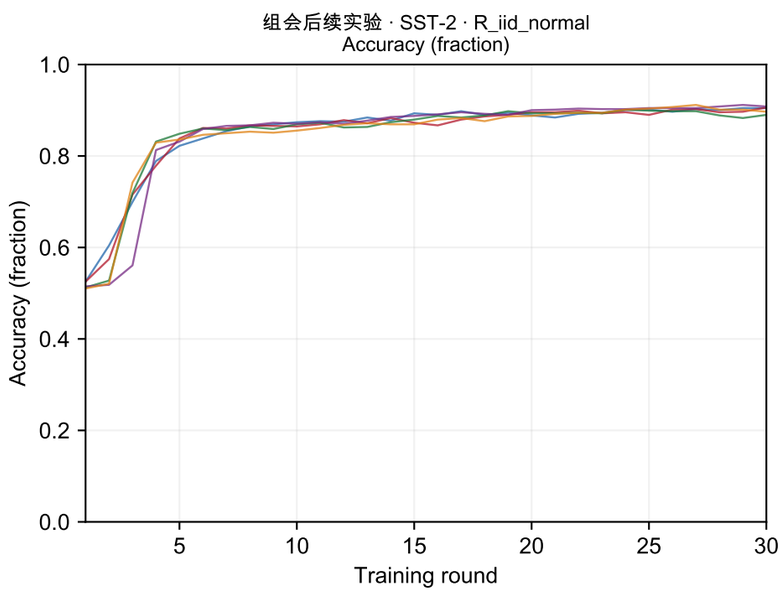


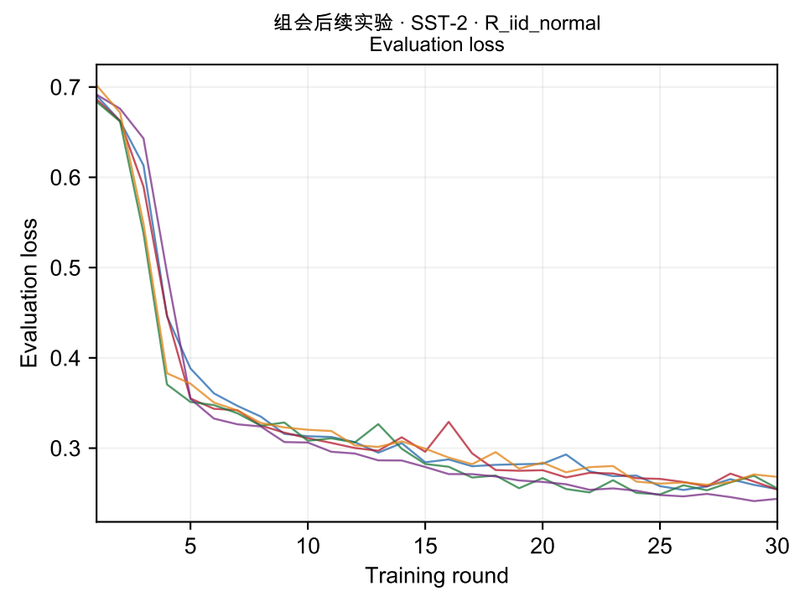
Full round trajectories · methods 31–32 of 45

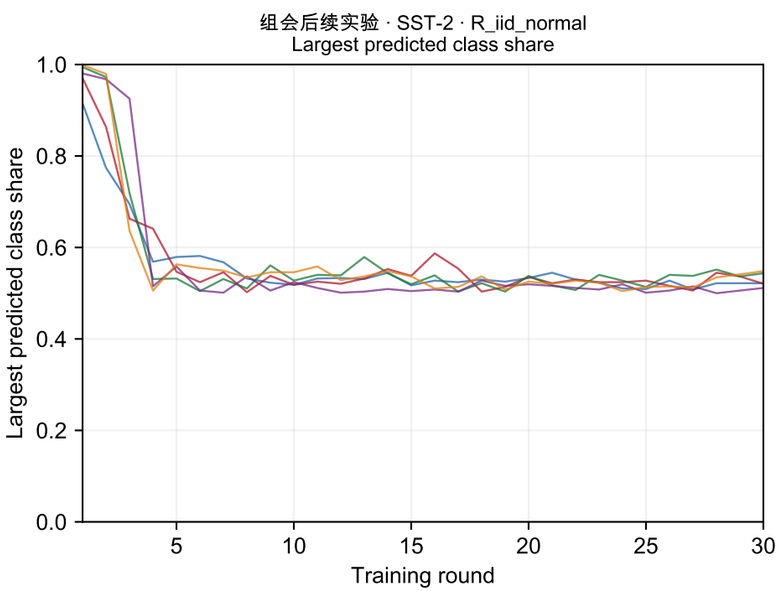


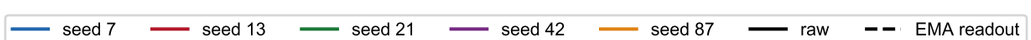




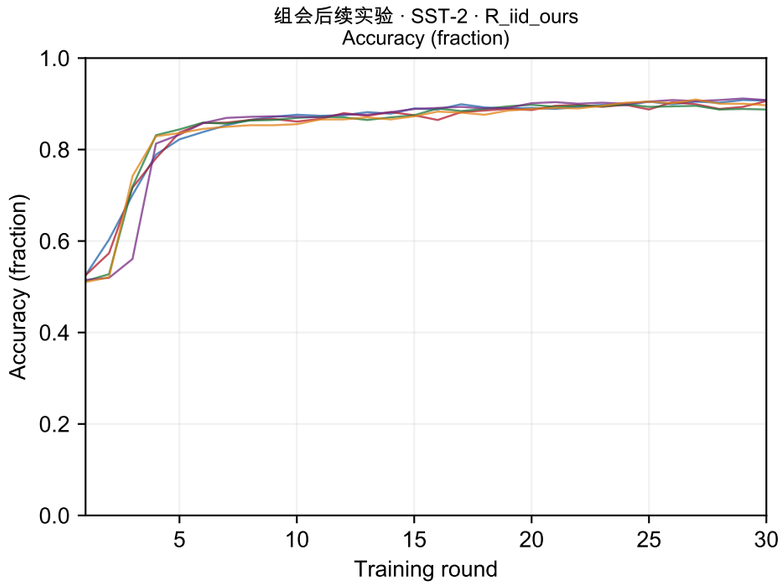


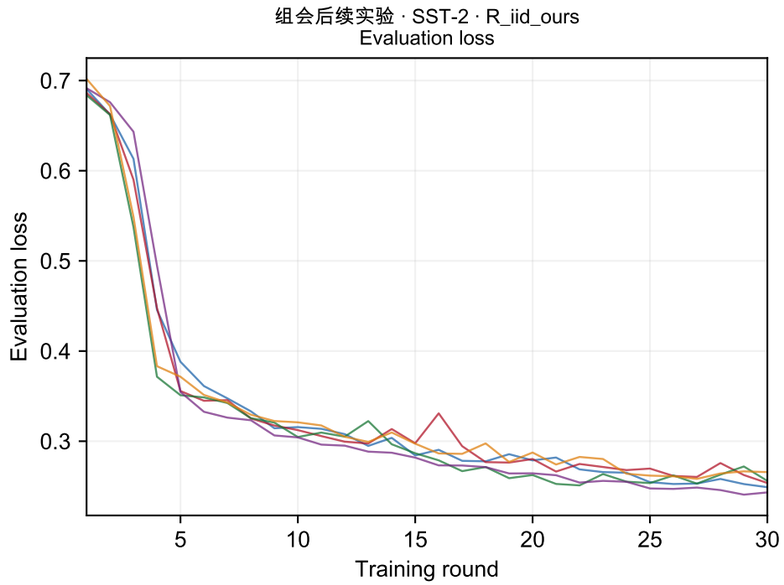


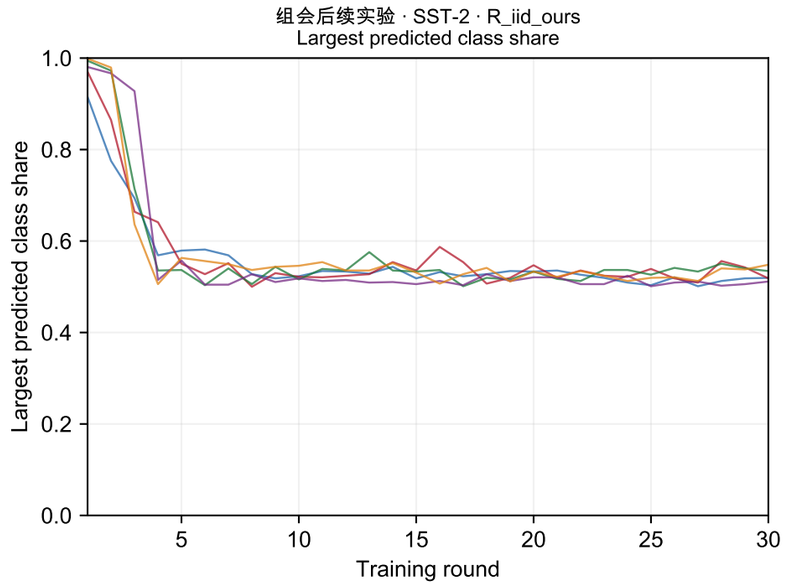


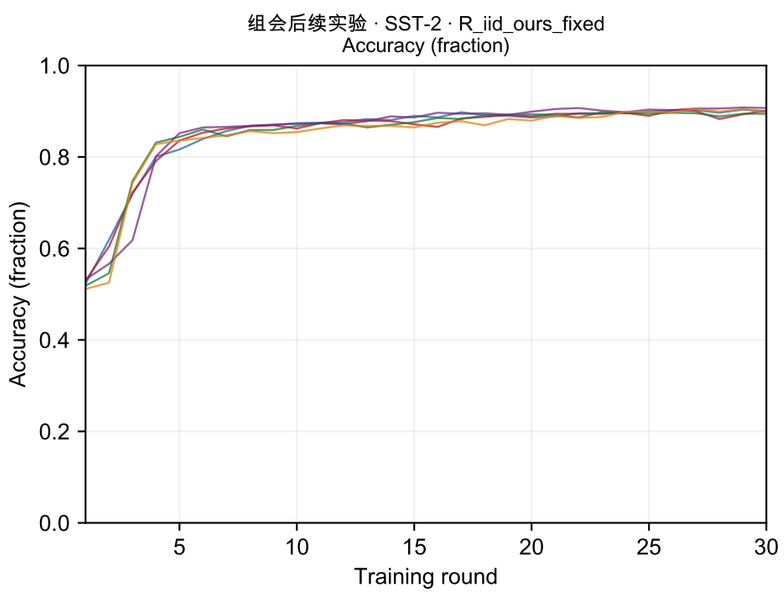


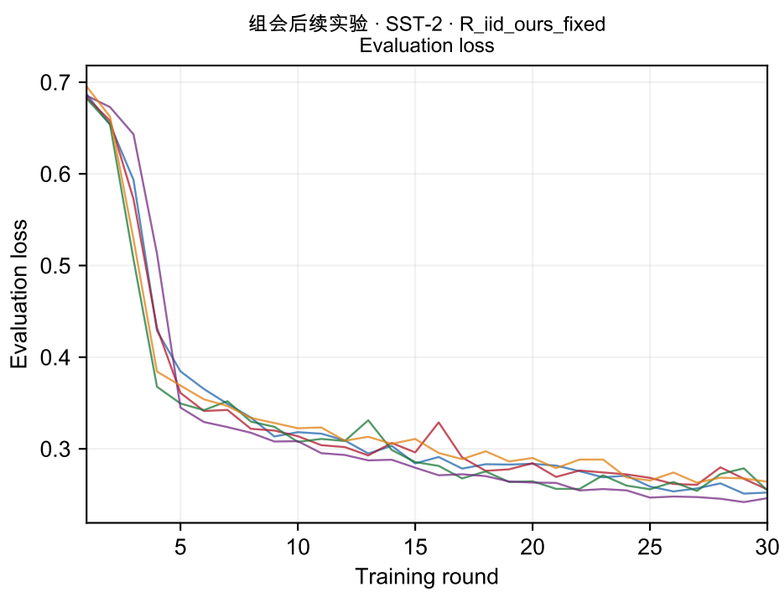
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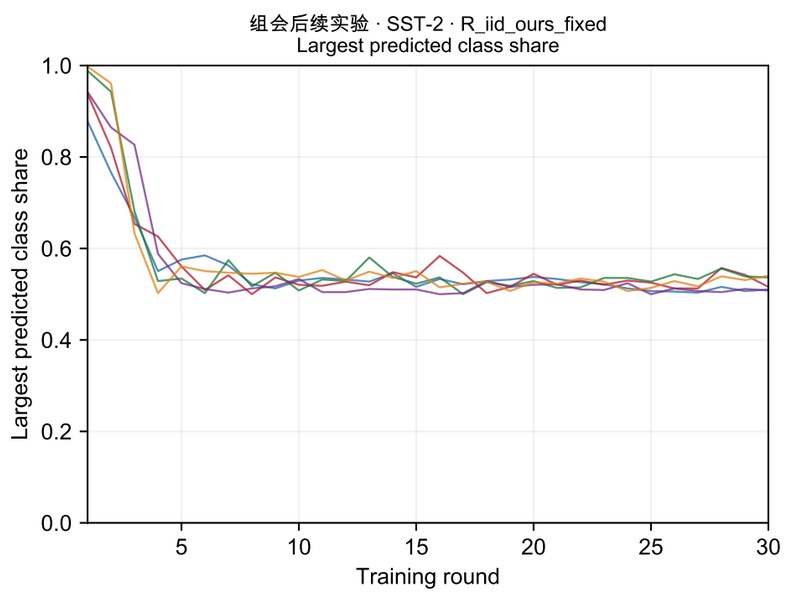


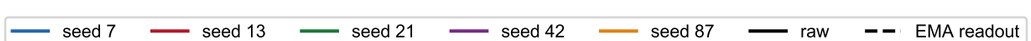




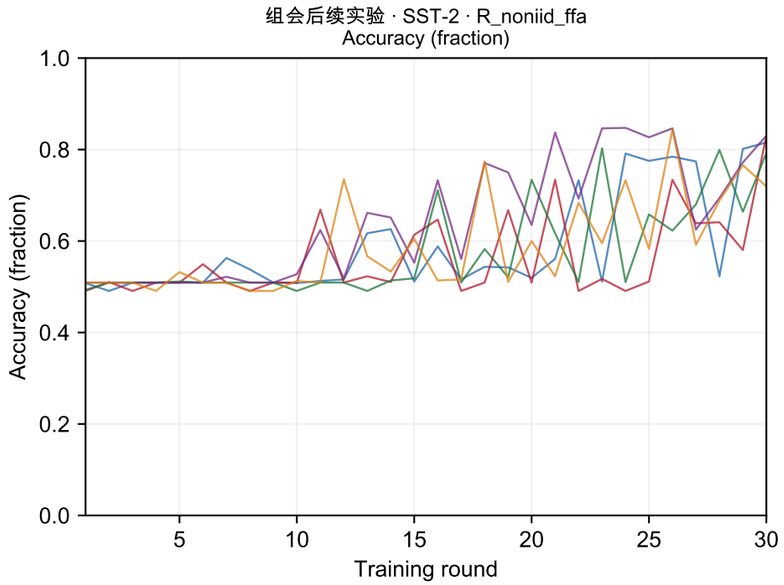


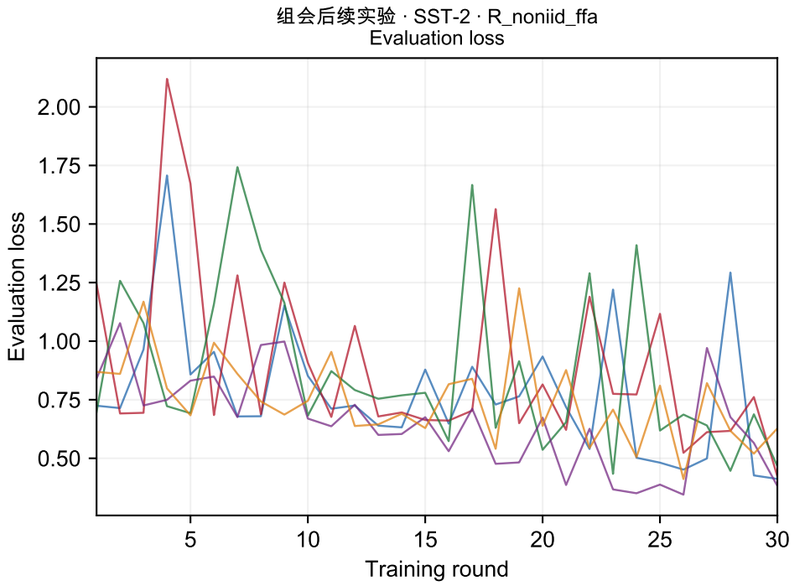


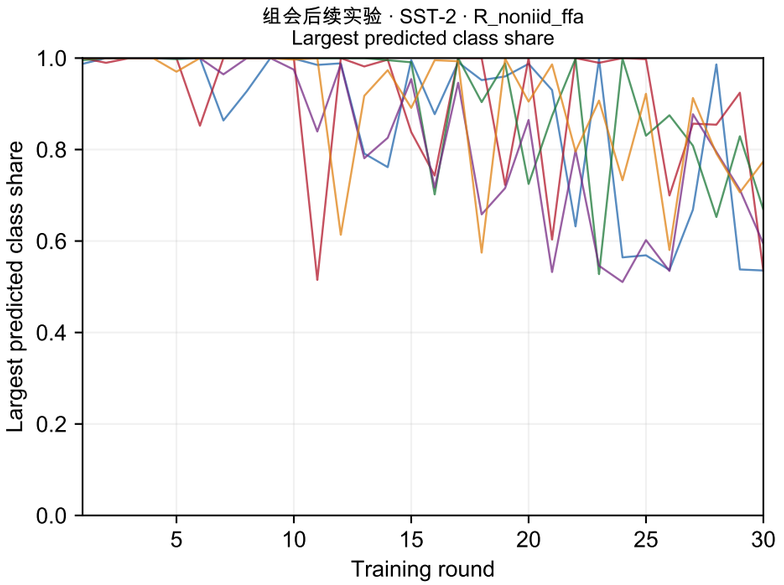


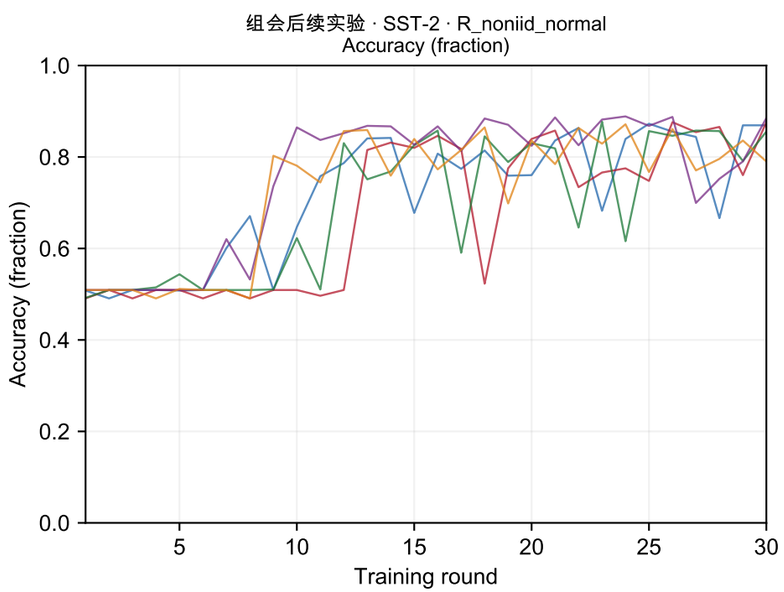


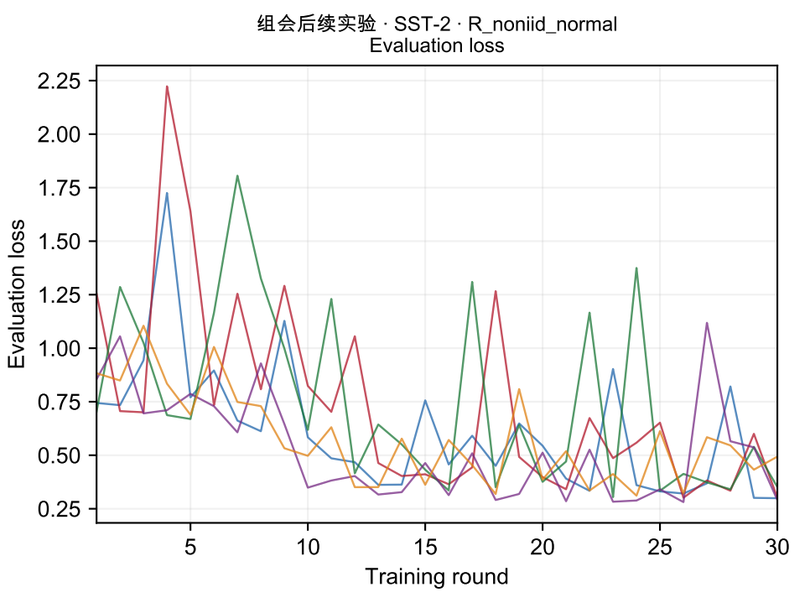
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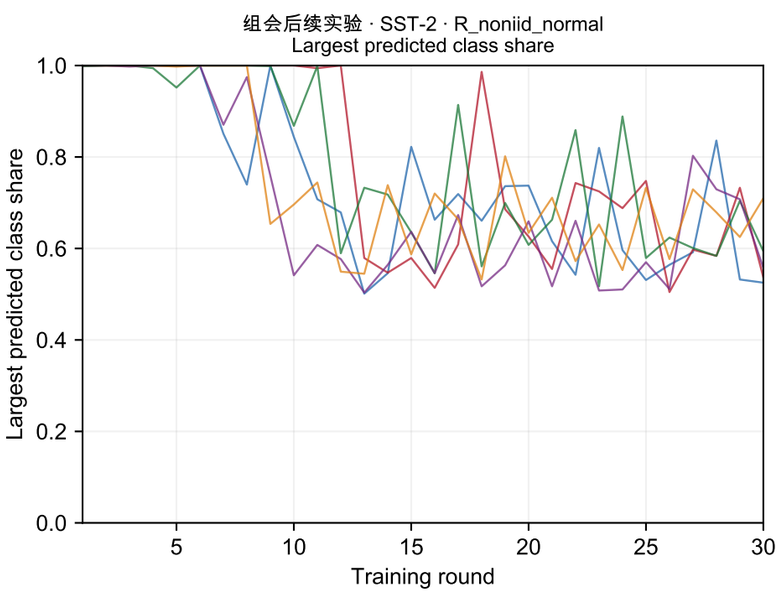


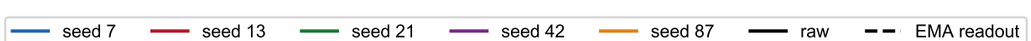




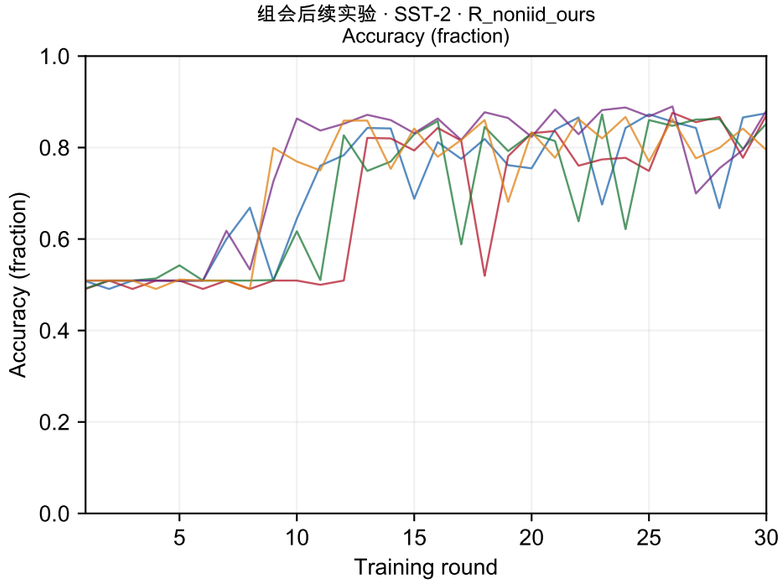


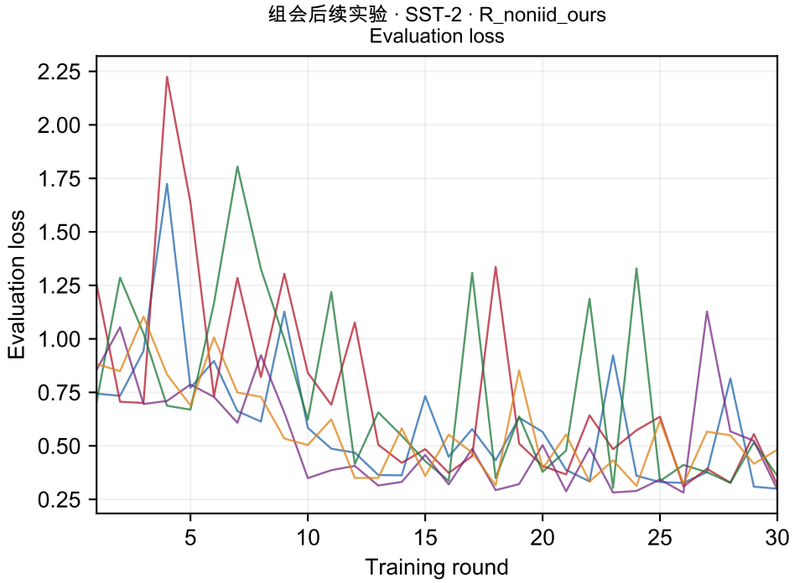


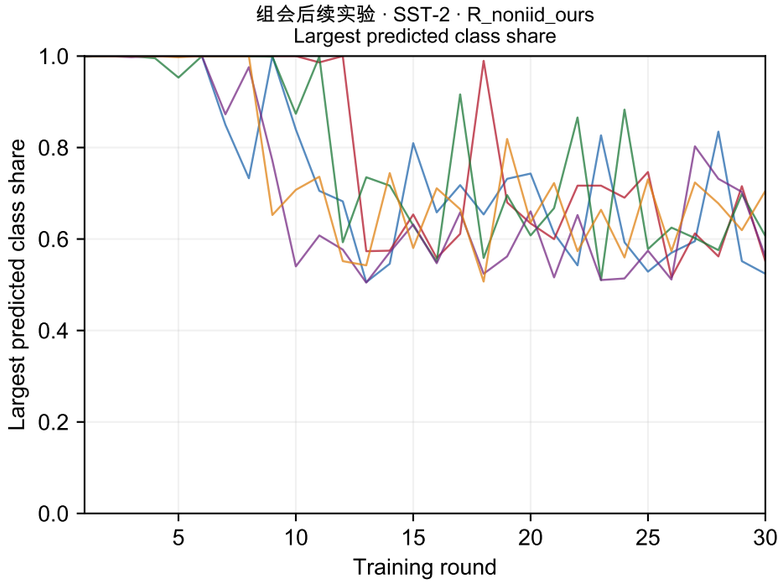


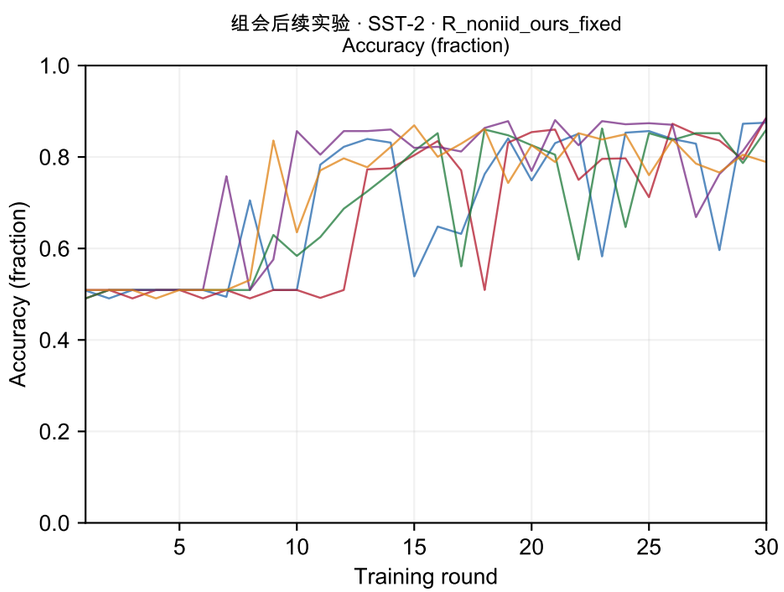


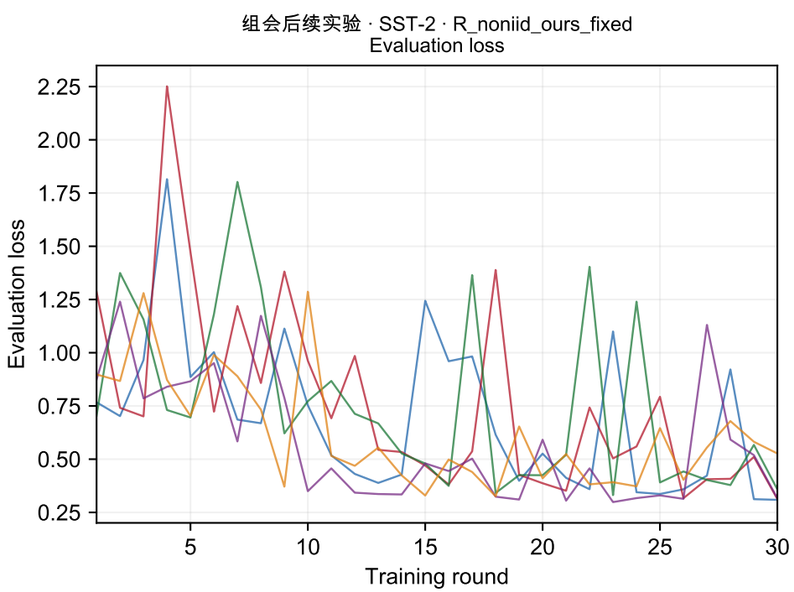
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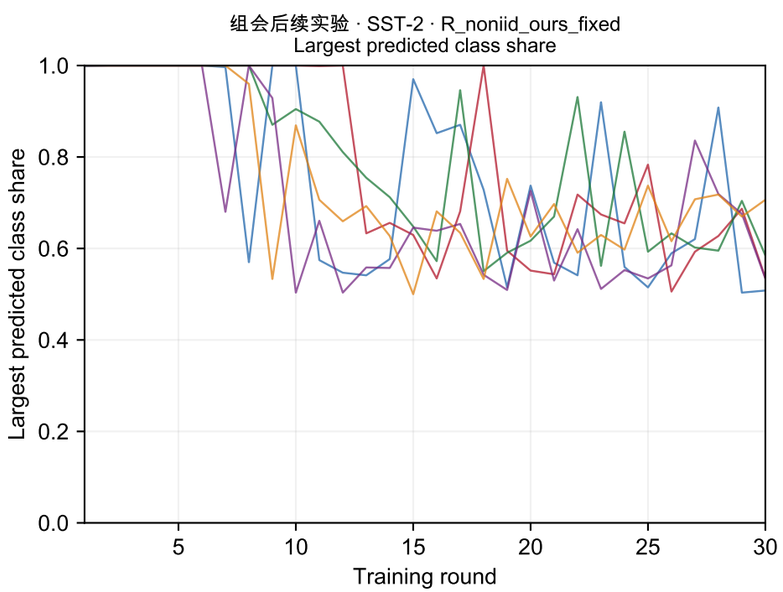


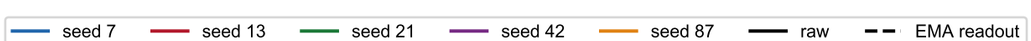




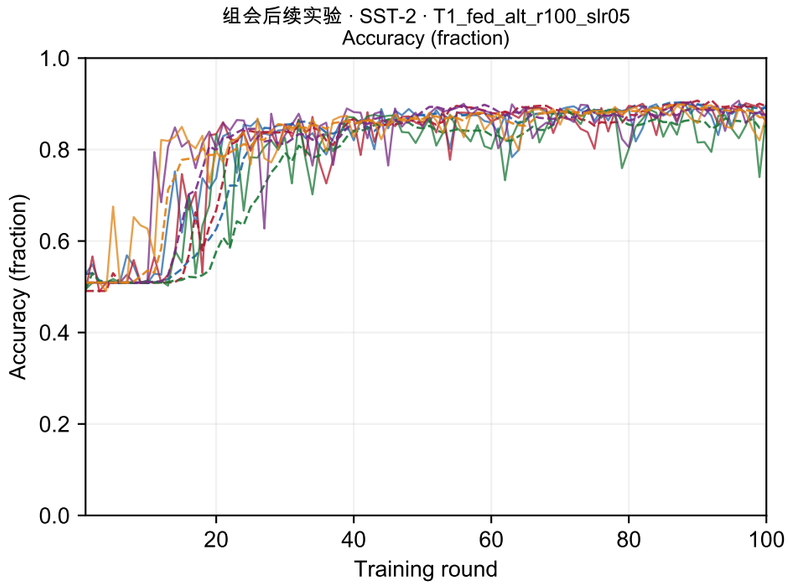


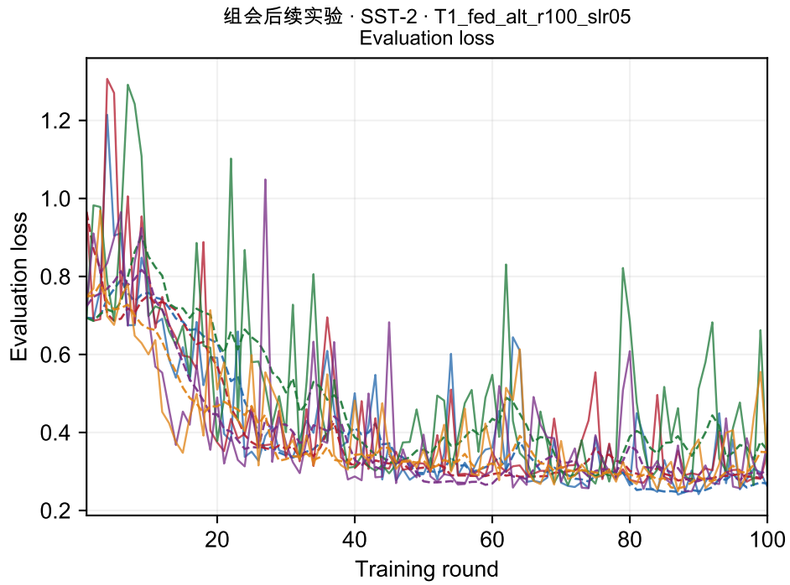


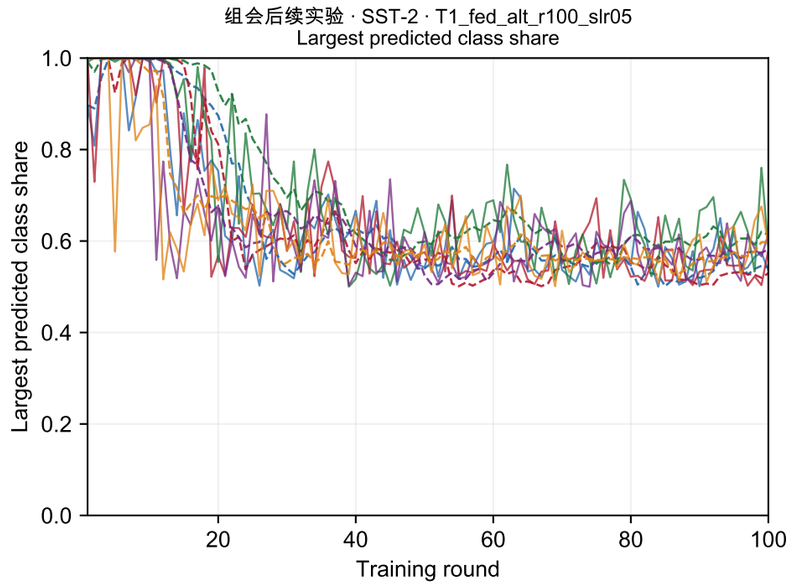


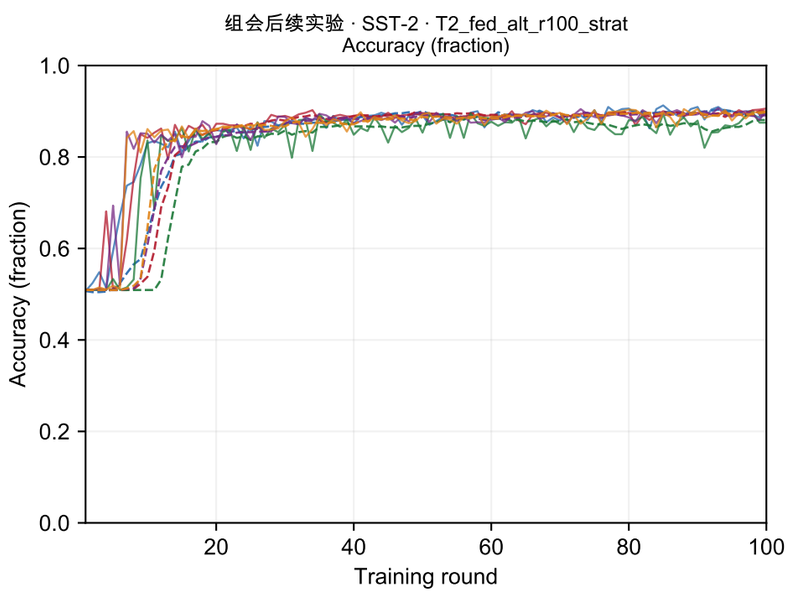


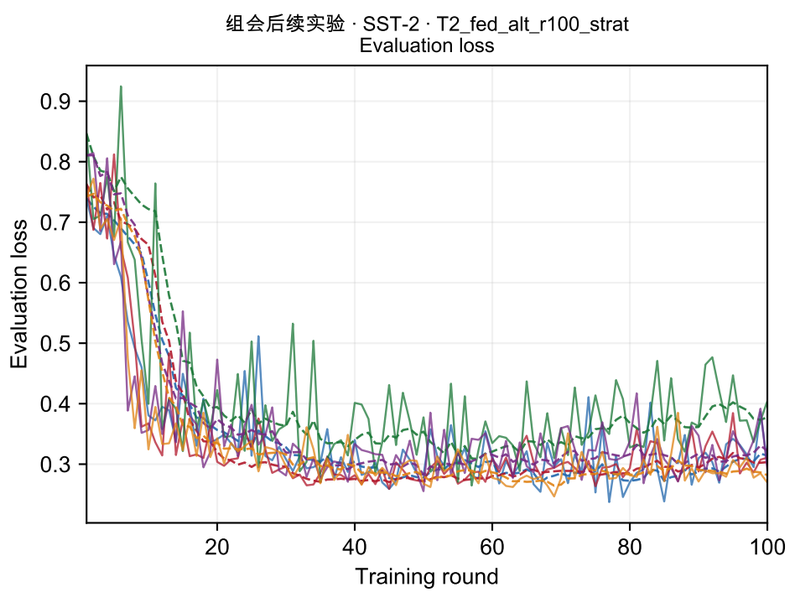
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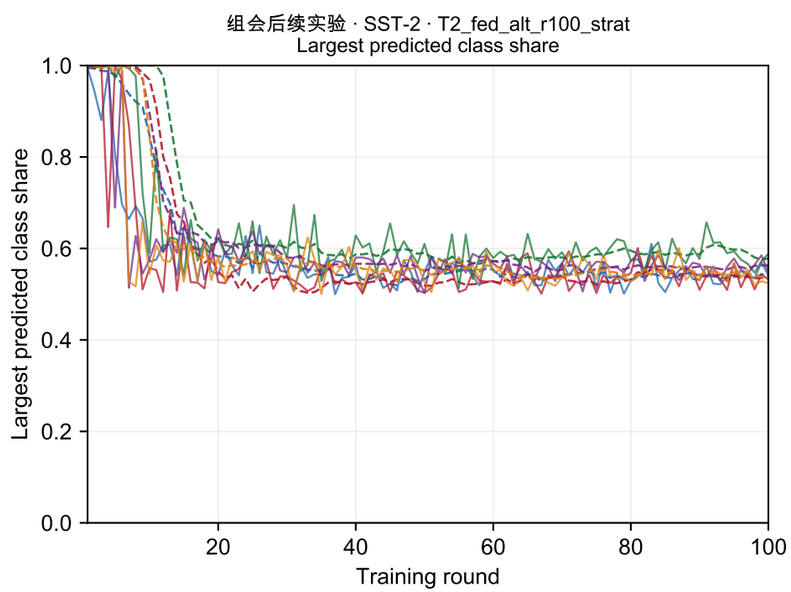


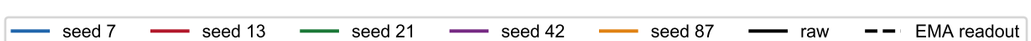




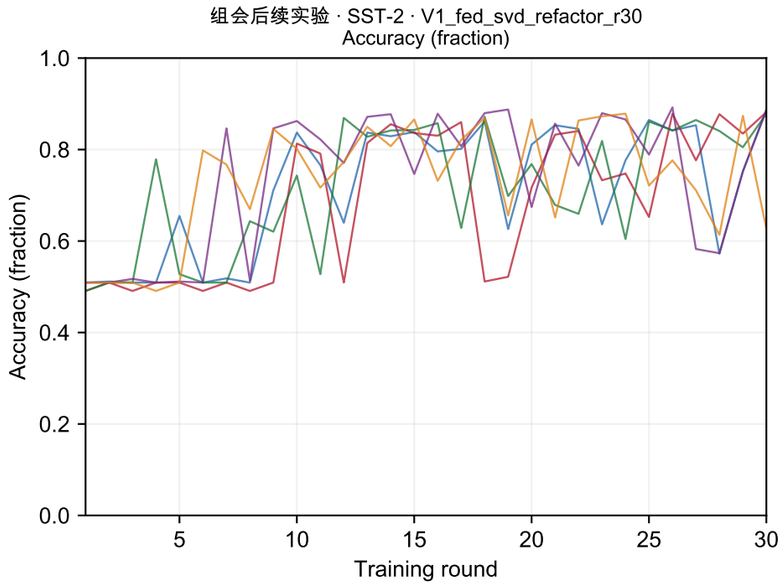


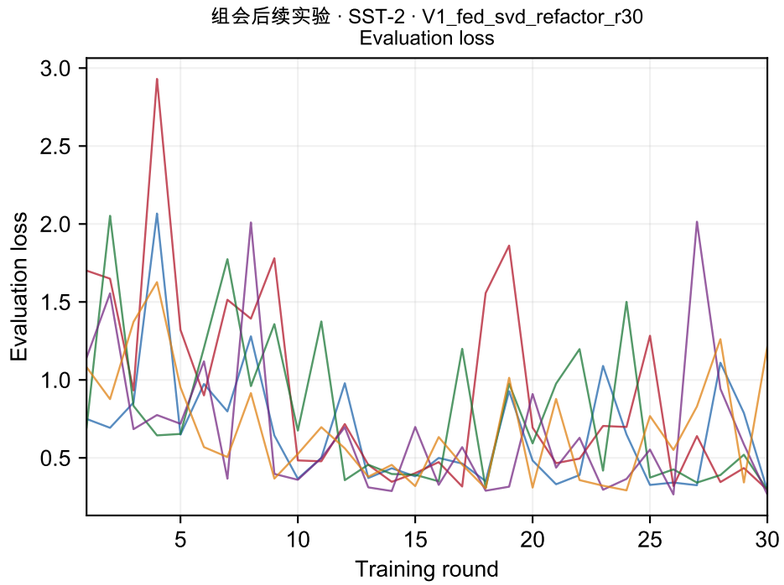


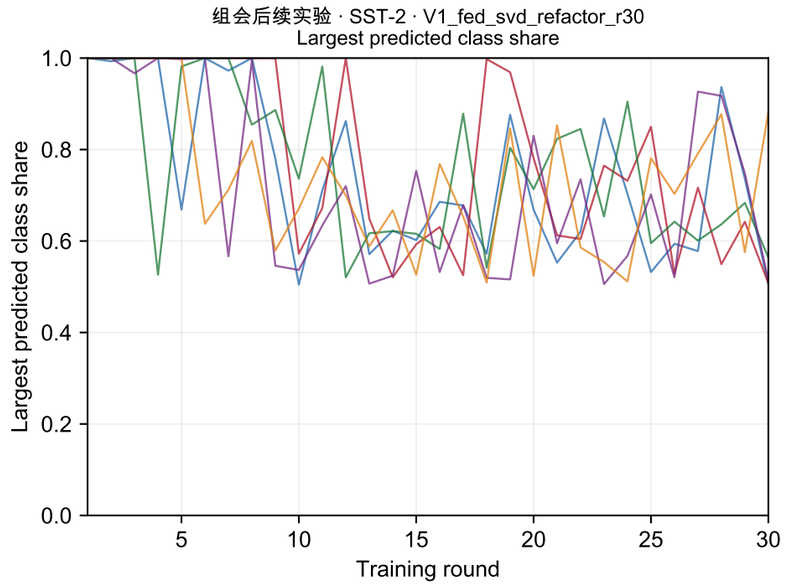


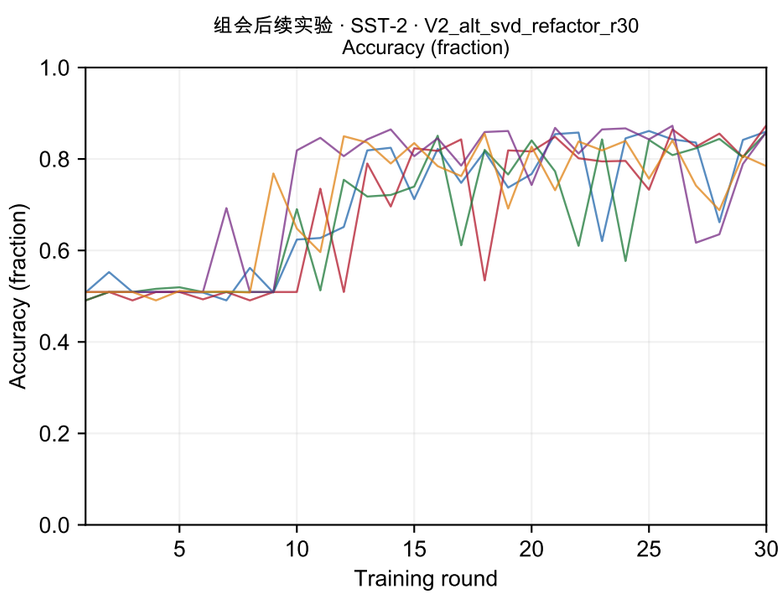


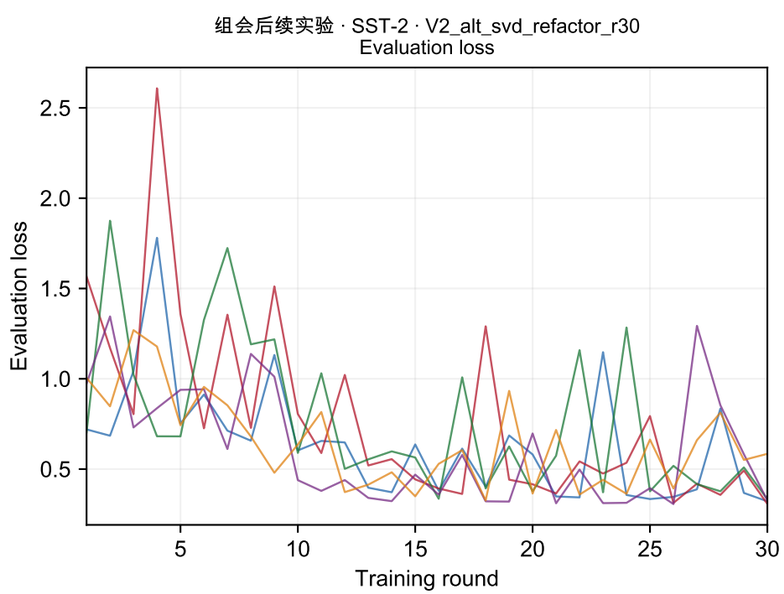
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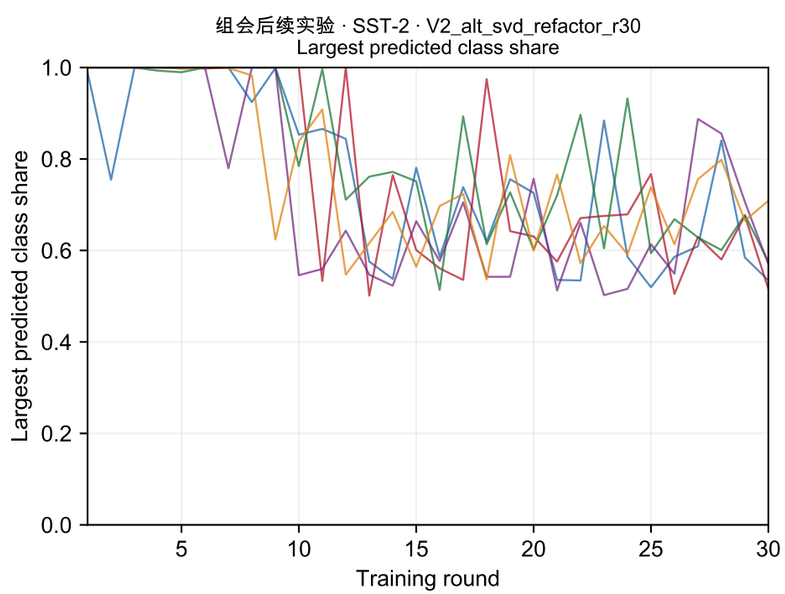


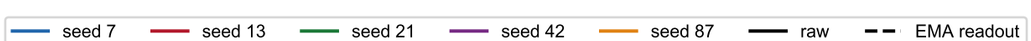




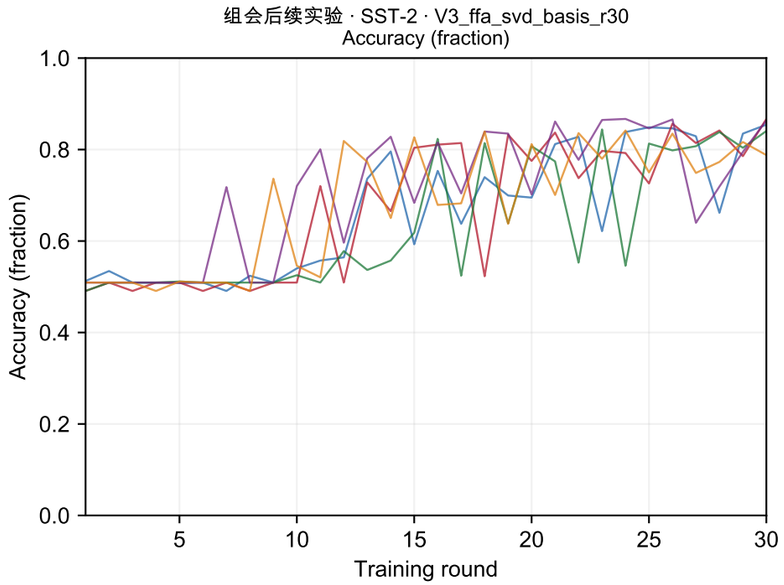


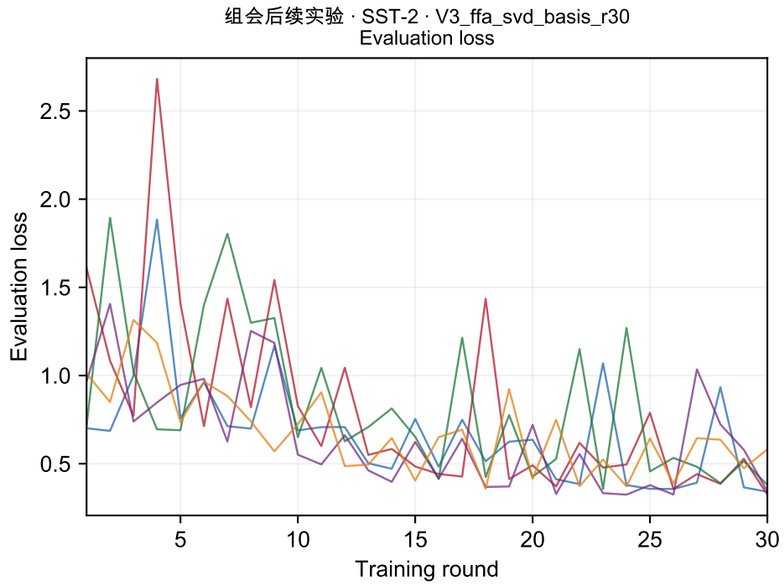


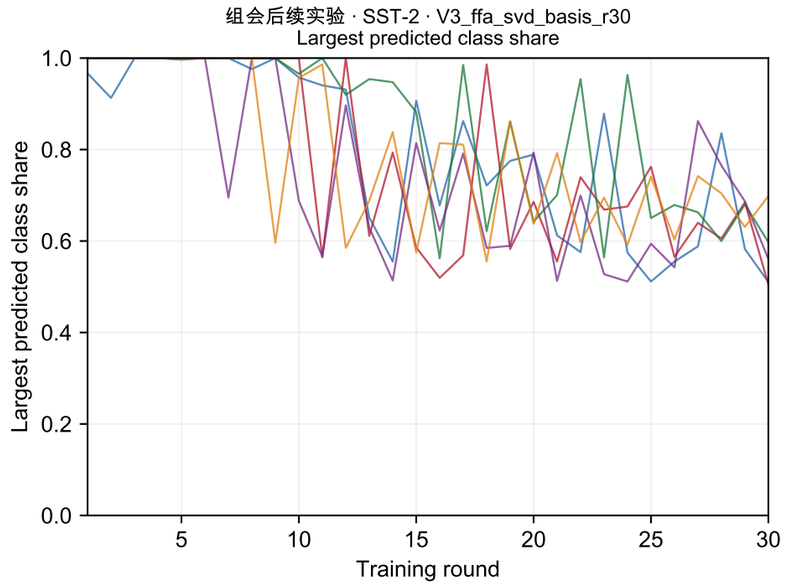


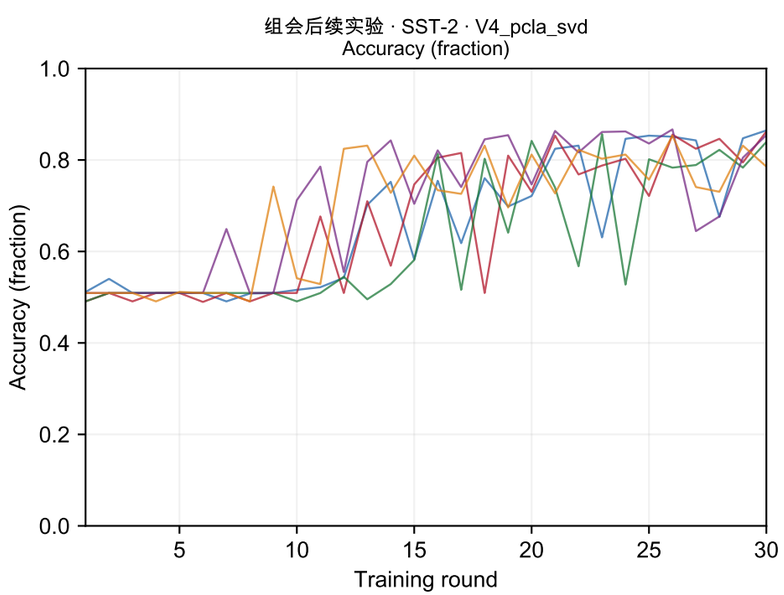


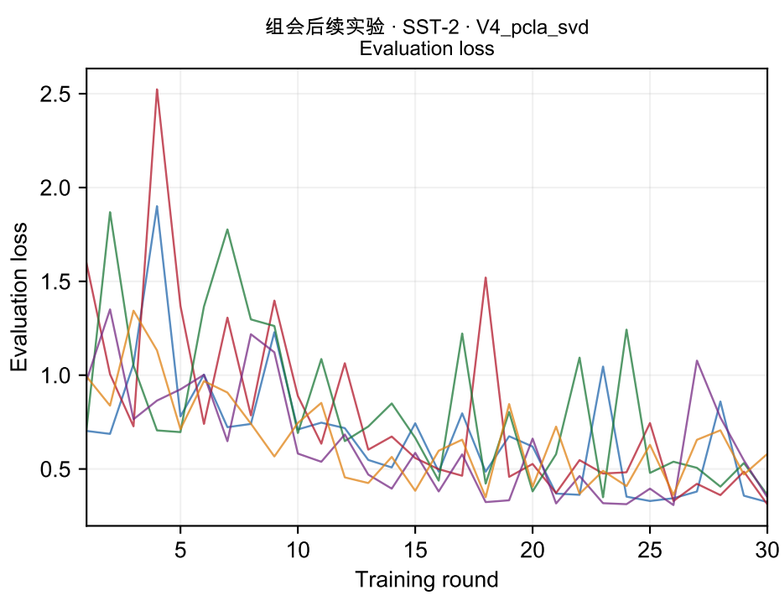
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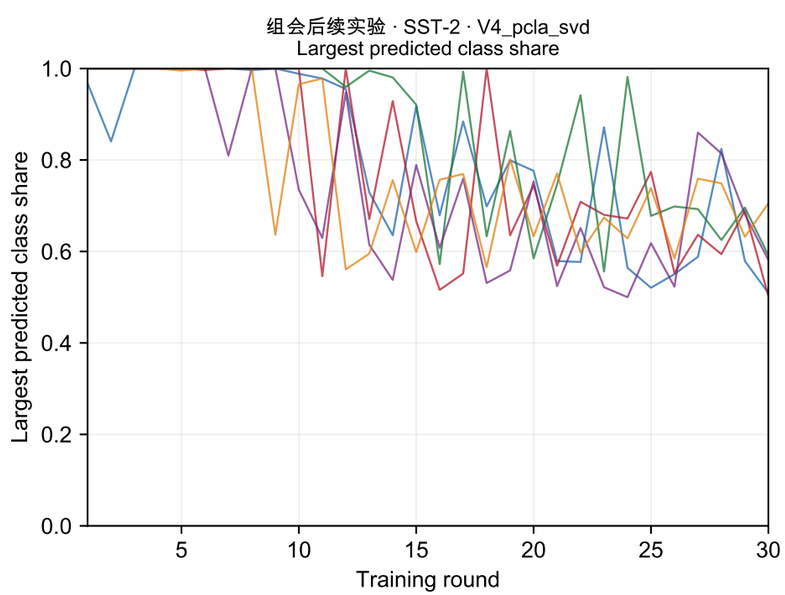


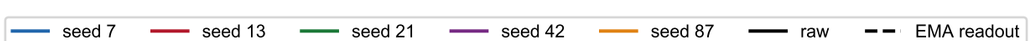












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